SMART REAL ESTATE ANALYTICS

PROJECT REPORT

Submitted to

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in partial fulfilment of the requirements for the award of

BACHELOR OF SCIENCE IN DATA SCIENCE

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in partial fulfilment of the requirements for the award of the degree **BACHELOR OF SCIENCE IN DATA SCIENCE , Bharathidasan University, Tiruchirappalli** during the sixth semester year 2025-2026.This is the original work of the candidate and was done under my guidance and supervision.

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VIVA-VOCE

VIVA-VOCE EXAMINATION

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ACKNOWLEDGEMENT

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ABSTRACT

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The real estate sector is one of the most economically significant industries in India, and accurate prediction of residential property prices remains a challenging task due to the highly multidimensional nature of the factors that influence valuation. Traditional appraisal methods that depend on human expertise and comparative market analysis are time-consuming, subjective, and difficult to scale across thousands of listings, making it essential to explore data-driven alternatives. This project presents a comprehensive machine learning-based house price prediction system built on a structured dataset of 2,844 residential property listings sourced from 12 major cities in Tamil Nadu, India, specifically Chennai, Coimbatore, Madurai, Trichy, Salem, Tirunelveli, Erode, Vellore, Thanjavur, Tiruppur, Hosur, and Nagercoil. The dataset contains 22 features including physical attributes such as area in square feet, number of bedrooms and bathrooms, floor level, property age, and parking availability, along with locational information including city, locality, latitude, and longitude, proximity-based features such as distances to city centers, schools, hospitals, metro stations, and highways, and additional attributes including furnished status, facing direction, amenities score, and year listed. The methodology follows a structured data science pipeline that begins with data collection and quality assessment, moves through thorough data cleaning including duplicate removal, missing value imputation, and outlier treatment, proceeds to exploratory data analysis and descriptive statistical profiling, applies feature engineering to create informative derived variables, and culminates in the training and evaluation of five machine learning regression algorithms namely Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost Gradient Boosting, and Support Vector Regression. Model performance is evaluated using Mean Absolute Error, Root Mean Squared Error, and the R-squared coefficient of determination, and hyperparameter tuning is performed through Grid Search Cross Validation to optimize the best-performing models. The experimental results demonstrate that ensemble methods, particularly XGBoost, significantly outperform linear baselines, achieving an R-squared score of 0.903 on the held-out test set, which means the model successfully explains over 90 percent of the variance in residential property prices across Tamil Nadu. The entire system is implemented in Python using Google Colab with libraries including Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and XGBoost, forming an end-to-end analytical pipeline that provides reliable, interpretable, and scalable price predictions for the South Indian residential real estate market.

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INTRODUCTION

1.INTRODUCTION

Overview

The real estate sector is one of the most economically significant industries in India, contributing nearly 7–8% of national GDP and providing large-scale employment across construction, brokerage, housing finance, and urban infrastructure. For most families, purchasing a home is the largest financial decision of their lifetime. Accurate property valuation at the time of purchase directly impacts long-term financial security, wealth accumulation, and credit risk management.

However, residential property valuation in India largely depends on traditional appraisal methods based on personal judgment and limited comparable transactions. These methods are subjective, inconsistent across regions, and difficult to scale across thousands of transactions. Buyers especially first-time homeowners - often lack access to objective price benchmarks and must rely on intermediaries. This creates information asymmetry and pricing inefficiencies in the market.

To address this challenge, this project introduces PropAura, a machine learning–based smart real estate analytics platform specifically designed for the Tamil Nadu residential market. By leveraging historical property listing data and advanced predictive algorithms, the system generates consistent, objective, and data-driven price estimates. The platform is deployed as an interactive web application accessible through a standard browser, enabling practical real-world usage.

Background

The foundation of quantitative property valuation lies in the hedonic pricing model proposed in 1974, which decomposes a property's value into the monetary contribution of its characteristics such as built-up area, number of bedrooms, age, and location. Early automated valuation models relied on linear regression techniques, which assumed simple linear relationships between features and price.

However, real estate markets exhibit complex non-linear and interaction effects. For example, the value of an additional bedroom depends on total area, location premiums vary across cities, and infrastructure proximity affects pricing differently in central and peripheral areas. To capture these complexities, modern machine learning methods such as Random Forest and XGBoost have been widely adopted due to their superior predictive performance and ability to model non-linear patterns.

Despite international advancements, machine learning-based valuation models tailored specifically to the Tamil Nadu market remain limited. The state exhibits distinct pricing dynamics, including strong variation between Chennai and secondary cities, infrastructure-driven micro-market differences, and diversity across property types. This

project fills that gap by developing a state-specific predictive system trained exclusively on Tamil Nadu data.

Motivation

Currently, property buyers in Tamil Nadu do not have access to objective price benchmarking tools. Online portals show listed prices but do not indicate whether they are fair. This lack of transparency increases financial risk, especially for first-time buyers.

Banks also require accurate valuations for mortgage risk assessment. Overvaluation increases credit risk, while undervaluation restricts loan access. A transparent, data-driven system improves fairness and supports informed decisions.

The dataset used includes 2,844 residential listings across twelve Tamil Nadu cities, with prices ranging approximately from ₹25.7 lakhs in Nagercoil to ₹61.4 lakhs in Chennai. It covers apartments, villas, independent houses, and plots.

Objectives

The primary objective of this project is to develop a machine learning model capable of accurately predicting residential property prices across twelve cities in Tamil Nadu, achieving performance superior to traditional linear regression methods. To accomplish this, the project first focuses on systematic data preprocessing and cleaning to ensure a high-quality, analysis-ready dataset. This is followed by comprehensive exploratory data analysis to understand price distributions, inter-city variations, and relationships between property features and prices. The project then involves training and evaluating an XGBoost model, with performance measured using Mean Absolute Error (MAE) and the R^2 coefficient of determination. To enhance transparency and user trust, SHAP-based explainability is integrated to provide feature-level insights for each prediction. Finally, the trained model is deployed as an interactive web application using Streamlit, supported by well-designed dashboards and visualisations to effectively communicate insights to both technical and non-technical users.

Scope of the Project

The project focuses on four property types—apartments, villas, independent houses, and plots—across twelve Tamil Nadu cities. It covers the complete machine learning pipeline from preprocessing to deployment using Python, Google Colab, and Streamlit.

Beyond price prediction, PropAura includes modules such as Smart Dashboard, Geo Intelligence, Investment Advisor, EMI Calculator, Property Comparator, and Blueprint Generator to support market analysis and financial planning.

Future improvements like real-time data integration and continuous retraining are outside the current scope.

SYSTEM ANALYSIS

2.SYSTEM ANALYSIS

System Analysis is the detailed study of the current real estate process and identifying its weaknesses in order to design a smarter, AI-driven solution. In this project, the system analysis focuses on traditional real estate practices and how they can be transformed into a Smart Real Estate System using data science and machine learning.

2.1 EXISTING MODELS

Overview

The existing real estate system primarily operates through real estate brokers, property agents, and online listing platforms. In most regions, property valuation is based on human experience rather than analytical computation. When a seller wants to determine the price of a property, they typically consult a broker who estimates the value by observing the location, infrastructure development, number of rooms, total area, and nearby property sales. These estimations are largely dependent on personal market knowledge rather than structured data analysis. Even though online platforms display property prices, those values are usually based on seller inputs and not on predictive algorithms.

The traditional system lacks an intelligent mechanism that can analyze historical housing data to determine accurate pricing. As a result, property valuation becomes subjective and varies from one agent to another. Buyers often depend on negotiation skills and market assumptions rather than data-backed insights.

Functioning of the Existing System

In the current system, the pricing process begins when a property owner contacts a broker or lists the property on a real estate website. The broker evaluates the property by physically visiting the location and comparing it with recently sold properties in the same area. The price is then suggested based on market trends, demand conditions, and the broker's previous experience. Buyers review multiple listings and manually compare prices before making a decision. The entire process depends heavily on manual comparison and human judgment.

There is no automated system that analyzes large volumes of historical housing data to detect hidden patterns or correlations. The absence of predictive modeling means that pricing decisions are reactive rather than analytical. The system does not dynamically adjust to new market data, which limits its reliability in rapidly changing markets.

Limitations of the Existing System

The traditional real estate system suffers from several limitations that reduce its efficiency and transparency. One of the major issues is the presence of human bias in pricing decisions. Since brokers rely on personal knowledge and negotiation strategies, different agents may quote different prices for the same property. This inconsistency creates confusion among buyers and sellers.

Another limitation is the absence of data analytics and statistical evaluation. The system does not analyze correlations between features such as area, number of bedrooms, bathrooms, and price trends. Without analytical insights, it becomes difficult to understand which factors truly influence property value. Additionally, there is no predictive capability to estimate future price growth or return on investment, which is crucial for investors.

The existing system also lacks visualization tools that can help users understand market behaviour. Buyers cannot view graphical representations of price distribution, feature importance, or location-based comparisons. This makes the decision-making process less informed and more dependent on assumptions.

Need for a Smart System

Due to the challenges identified in the traditional approach, there is a strong need for a smart, data-driven real estate system. The growing availability of housing datasets and advancements in machine learning create an opportunity to automate and improve property valuation. A smart system should be capable of analyzing historical data, identifying influential features, predicting accurate prices, and providing transparent insights. This requirement forms the foundation for the proposed Smart Real Estate System.

2.2 PROPOSED SYSTEM

The proposed system**, PropAura** – Smart Real Estate Analytics Platform, introduces a comprehensive machine learning–based solution for residential property price prediction across twelve Tamil Nadu cities. Unlike traditional appraisal methods, this system leverages advanced algorithms, structured data processing, and interactive visualization to deliver accurate, consistent, and explainable price estimates.

1. Machine Learning-Based Prediction

At the core of the proposed system is the XGBoost (Extreme Gradient Boosting) algorithm. XGBoost is an advanced ensemble learning technique that builds multiple decision trees sequentially, where each tree corrects the errors of the previous one. This approach enables the model to capture complex non-linear patterns and feature interactions present in real estate markets.

The model is trained using 2,844 residential property listings containing features such as built-up area, number of bedrooms, bathrooms, city, property type, furnishing status, and amenities. The dataset is divided into training and testing sets using an 80:20 ratio to ensure unbiased evaluation. Performance metrics such as Mean Absolute Error (MAE) and R^2 score are used to measure predictive accuracy.

Compared to traditional linear regression, XGBoost provides significantly improved performance by handling non-linearity, reducing overfitting through regularization, and automatically identifying feature importance.

2. Data Preprocessing and Feature Engineering

The proposed system includes a structured data preprocessing pipeline. This involves handling missing values, removing duplicates, correcting inconsistent formats, encoding categorical variables, and scaling numerical features where necessary. Feature selection techniques are applied to identify the most relevant predictors affecting property price.

By preparing a clean and standardized dataset, the system ensures higher model reliability and stability.

3. Explainable Artificial Intelligence (XAI) Integration

One of the key innovations of the proposed system is the integration of SHAP (SHapley Additive exPlanations). SHAP provides both global and local interpretability.

- Global explainability identifies which features most influence overall pricing trends.
- Local explainability shows how each feature contributes to an individual property's predicted price.

This transparency improves user trust and allows buyers, sellers, and lenders to understand the reasoning behind each valuation.

4. Interactive Web Application Deployment

The trained model is deployed using the Streamlit framework, transforming the predictive system into a fully interactive web application. Users can input property details such as city, property type, area, and number of bedrooms to receive instant price predictions along with feature contribution explanations.The web interface is designed to be simple and user-friendly, making advanced machine learning accessible even to non-technical users.

5. Multi-Module Analytics Platform

Unlike basic prediction tools, PropAura functions as a comprehensive real estate analytics ecosystem. The system includes:

- Smart Dashboard for market trend visualization
- Geo Intelligence Module for location-based insights
- Investment Advisor for ROI-based recommendations
- EMI Calculator for loan planning
- Property Comparator for side-by-side comparison
- Blueprint Generator for architectural layout support

These modules collectively provide a holistic decision-support system for property buyers and investors.

6. Scalability and Future Readiness

The proposed system is scalable and can be extended to additional cities or integrated with real-time data sources. Its modular architecture allows continuous model retraining and feature updates as new market data becomes available. This ensures adaptability to dynamic real estate market conditions.

Advantages of the Proposed System

1. Higher prediction accuracy compared to traditional linear models.
2. Ability to capture complex non-linear relationships.
3. Transparent and explainable predictions using SHAP.
4. Scalable and automated valuation process.
5. User-friendly web deployment for real-world accessibility.

In conclusion, the proposed system overcomes the major limitations of traditional valuation methods by introducing a data-driven, accurate, explainable, and scalable machine learning framework. It transforms residential property valuation from a subjective manual process into a reliable and intelligent analytical system tailored for the Tamil Nadu market.

PROBLEM DESCRIPTION

3.PROBLEM DESCRIPTION

Background

The Tamil Nadu residential real estate market is one of the fastest-growing property sectors in South India, covering 12 major cities — Chennai, Coimbatore, Madurai, Trichy, Salem, Tirunelveli, Erode, Vellore, Thanjavur, Tiruppur, Hosur, and Nagercoil. Based on the dataset used in this project, which contains 2,844 real property listings, prices range from Rs.8.86 lakhs to Rs.4.57 crores with an average market value of Rs.56.03 lakhs. The market covers four property types — Apartment, Villa, Independent House, and Plot — with built-up areas ranging from 450 sqft to 4,983 sqft.

Despite this scale, a major problem exists: there is no accurate, data-driven, and accessible tool to determine the fair market value of a residential property in Tamil Nadu. Buyers rely on biased agent quotes, outdated government circle rates, or expensive manual appraisals that take 3 to 7 days and still produce valuations that differ by 10 to 25% between two appraisers. This absence of a reliable valuation system causes financial losses for buyers, missed revenue for sellers, risk exposure for lenders, and overall inefficiency across the entire Tamil Nadu real estate market. This project, PropAura, is built to solve this problem using machine learning.

Information Asymmetry

One of the most serious problems in the Tamil Nadu property market is information asymmetry — a situation where sellers, developers, and agents know far more about the true market value of a property than buyers do. Sellers are aware of recent transaction prices in their locality. Developers have advance knowledge of upcoming infrastructure projects such as metro extensions or IT park announcements that will increase property values before the information becomes public. Local agents share actual transaction prices within their private networks, information that is never published on any public portal.

Buyers — especially first-time homebuyers, those purchasing in a new city, or NRI buyers purchasing remotely from abroad — have no access to this information. Without a reliable and independent price reference, buyers cannot negotiate effectively and are forced to either trust the agent's quoted price or spend weeks manually visiting multiple properties to gather informal price data. This process is impractical for working professionals and impossible for remote buyers. Research shows that buyers with access to reliable market data negotiate prices 7 to 12% lower than uninformed buyers for identical properties. On an average Tamil Nadu property worth Rs.56.03 lakhs, this means uninformed buyers overpay by approximately Rs.3.92 to Rs.6.72 lakhs — a significant financial loss that a data-driven valuation tool like PropAura can prevent.

Inability to Capture Non-linear Relationships

Property prices do not follow simple straight-line patterns. In this dataset, area ranges from 450 sqft to 4,983 sqft and prices span Rs.8.86 lakhs to Rs.4.57 crores — and the relationship between them is not linear. The value added by each additional square foot decreases as the property gets larger. A brand-new property commands a sharp price premium, but a 10-year-old property depreciates gradually, while a 30-year-old heritage property may actually appreciate due to scarcity. Ground floor carries no premium, floors 1–10 carry a progressive premium, and very high floors may carry a discount due to elevator dependency. Existing tools in Tamil Nadu use simple linear regression that assumes every feature adds a fixed, constant amount to the price — causing them to systematically undervalue luxury properties and overvalue budget ones. XGBoost, used in PropAura, captures all of these complex curves and interactions natively.

Lack of Geospatial Intelligence

Location is the most important determinant of property value, yet existing tools represent it only as a city name or locality label — a critically crude approximation. This dataset includes five GPS-based proximity features computed using the Haversine formula: distance to city center, distance to nearest school, distance to nearest hospital, distance to nearest metro station, and distance to nearest highway. These features reveal price differences that a city label completely hides. Within the same city, a property 1 km from a metro station can be worth 20–35% more than an identical property 5 km away. Proximity to a reputed school adds an 8–15% premium. A property 500 metres from a highway carries a noise and pollution discount, while one 1–3 km away enjoys connectivity without the drawbacks. PropAura is the first Tamil Nadu valuation system to include all five geospatial distance features as direct model inputs.

Absence of Explainability

Even where automated price estimates exist on portals like MagicBricks or 99acres, they function as complete black boxes. The system outputs a price — for example, Rs.68 lakhs — with no explanation of how that number was calculated or which features drove it. The buyer cannot know whether the estimate is primarily driven by the property's large area, its proximity to a metro station, its fully furnished status, or its new construction age. Without this information, buyers cannot negotiate using specific data-backed arguments, sellers cannot identify which upgrades would add the most value, and lenders cannot document which features justify the valuation for their loan approval records.

The consequence is a severe trust deficit. Research shows that approximately 74% of Indian home buyers do not trust automated valuation tools because they cannot understand the calculation behind the price. This distrust forces buyers back to relying on biased agent opinions — the exact problem automated tools were meant to solve. A black-box model that produces a number without explanation is, in practice, useless to the average buyer who has no way to verify or challenge it.

PropAura solves this completely using SHAP — SHapley Additive exPlanations — a mathematically rigorous framework based on game theory that breaks down every single prediction into individual feature contributions. For any property, PropAura does not just show the predicted price of Rs.68 lakhs — it shows that the property's area contributes +Rs.12 lakhs, the city location contributes +Rs.8 lakhs, the proximity to a metro station contributes +Rs.5 lakhs, the 12-year age reduces value by -Rs.4 lakhs, and the unfurnished status reduces value by -Rs.3 lakhs. This level of feature-level transparency transforms PropAura from a black-box estimator into a genuine advisory tool that empowers every stakeholder to make informed decisions.

No Financial Planning Support

Knowing the market price of a property is only the first step. A buyer also needs to know: Can I afford this? What will my monthly EMI be? What will this property be worth in 10 years? Is buying in Chennai a better investment than Coimbatore for my budget? None of these questions are answered by any existing Tamil Nadu real estate tool. Buyers are forced to use a bank's mortgage calculator separately, manually compare city appreciation rates from news articles, and guess at rental yields — a fragmented and error-prone process. PropAura integrates an EMI Calculator, an Investment Advisor with ROI projections, and a Property Comparator directly into the same platform as the price prediction engine.

Problem Statement

The Tamil Nadu residential property market — covering 2,844 listings across 12 cities, 4 property types (Apartment, Villa, Independent House, Plot), and a price range of Rs.8.86 lakhs to Rs.4.57 crores — lacks a machine learning system that can accurately predict property prices using non-linear modelling, incorporate real GPS-based geospatial features, explain every prediction at the feature level, cover all 12 cities in one unified model, and deliver these capabilities through a web interface accessible to non-technical users. PropAura is built to solve exactly this problem.

DATA COLLECTION

4.DATA COLLECTION

4.1 Dataset Description

The dataset used in this project is a structured tabular dataset containing 2,844 residential property listings from 12 major cities in Tamil Nadu, India. Each record in the dataset represents a single property listing and is described by 22 attributes spanning four categories: physical property characteristics, geographic location information, proximity-based accessibility metrics, and the target variable representing the total property price in Indian Rupees. The dataset was assembled from a combination of online real estate platforms, government transaction records, and geographic data sources, providing a comprehensive snapshot of the Tamil Nadu residential market across the period from 2016 to 2023.

Feature Description

The dataset contains the following 22 features. The locality feature records the specific neighbourhood or area name within the city, such as Tambaram or Anna Nagar in Chennai, providing fine-grained locational identification. The city feature records one of the 12 Tamil Nadu cities: Chennai, Coimbatore, Madurai, Trichy, Salem, Tirunelveli, Erode, Vellore, Thanjavur, Tiruppur, Hosur, or Nagercoil. The latitude and longitude features provide the precise geographic coordinates of the property in decimal degrees, enabling spatial analysis and computation of geographic distances.

The property_type feature categorizes each listing as one of four types: Apartment, Villa, Independent House, or Plot. Apartment refers to a unit within a multi-story residential building and is the most common type in the dataset. Villa refers to a standalone luxury residential property typically with a private garden or compound. Independent House refers to a standalone non-luxury residential property, typically a traditional bungalow or row house. Plot refers to a land parcel without any constructed structure.

The area_sqft feature records the built-up area of the property in square feet, ranging from 450 to 4,983 square feet across the dataset. The bedrooms and bathrooms features record the number of respective rooms. The floor feature records the floor number on which an apartment unit is located, while total_floors records the total number of floors in the building. The age_years feature records the age of the constructed property in years at the time of listing. The parking feature records the number of dedicated parking spaces available with the property. The furnished feature categorizes properties into three states: Furnished, Semi-Furnished, and Unfurnished. The facing feature records the cardinal or intercardinal direction the primary facade of the property faces, with eight possible values: North, South, East, West, North-East, North-West, South-East, and South-West.

The amenities_score feature is a composite numeric score on a scale of 0 to 10 representing the availability and quality of amenities within the property complex, including gym, swimming pool, clubhouse, security, power backup, and similar facilities. The five proximity features measure the straight-line distance in kilometers from the property to the nearest city center, nearest school, nearest hospital, nearest metro or suburban rail station, and nearest highway, respectively. The year_listed feature records the year in which the property was listed for sale, ranging from 2016 to 2023. Finally, price_inr is the target variable recording the total listed price of the property in Indian Rupees.

Dataset Statistics

The dataset exhibits the following key statistical characteristics. In terms of city distribution, Chennai contributes the most listings with 540 records, reflecting its status as the largest and most active property market in Tamil Nadu. Coimbatore follows with 360 records, Madurai and Trichy each with 270, Salem with 216, and the remaining cities — Tirunelveli, Erode, Vellore, Thanjavur, and Tiruppur — with 180 records each, while Hosur and Nagercoil each contribute 144 recordsThe four property types are distributed as follows: Apartments account for 1,528 records representing 53.7 percent of the dataset, Villas account for 552 records representing 19.4 percent, Independent Houses for 492 records representing 17.3 percent, and Plots for 272 records representing the remaining 9.6 percent. The predominance of apartments reflects the urbanization trend across Tamil Nadu's major cities, where multi-story residential development has become the dominant housing form.

The price variable ranges from a minimum of Rs.8,86,000 to a maximum of Rs.4,57,86,000, with a mean of approximately Rs.56.02 lakhs and a median of approximately Rs.41.84 lakhs. The significant gap between mean and median confirms the right-skewed nature of the price distribution, where a small number of high-value luxury properties pull the mean well above the median. Area ranges from 450 to 4,983 square feet with a mean of approximately 1,332 square feet. The amenities score ranges from 0.0 to 10.0 with a mean of approximately 5.02, indicating that the average property in the dataset offers a moderate amenity package. The dataset contains zero missing values across all 22 features, confirming its completeness and readiness for model training without any imputation required.

Regarding furnishing status, 942 properties are Furnished (33.1%), 1,046 are Semi-Furnished (36.8%), and 856 are Unfurnished (30.1%), showing a relatively balanced distribution across all three furnishing states.

4.2 Data Sources

The dataset for this project was assembled from three primary categories of sources, each contributing a distinct type of information to the final dataset.

Online Real Estate Platforms (via Kaggle)

The primary source of property listing data is the Kaggle open data platform, which hosts a curated Tamil Nadu residential property dataset representing an aggregation of listings from India's major real estate portals. The three main portals contributing to this dataset are MagicBricks, 99acres, and Housing.com, which together account for the vast majority of residential property listings in Tamil Nadu's organized real estate market.

MagicBricks (www.magicbricks.com) is one of India's largest real estate portals, operated by Times Internet Limited. It hosts verified property listings from across Tamil Nadu with detailed attributes including area, BHK configuration, floor level, furnishing status, and amenity information. MagicBricks listings contributed the structural and categorical feature data for a significant portion of the dataset's records.

99acres (www.99acres.com) is operated by Info Edge India Limited and is one of the most widely used property search portals in South India. It is particularly strong in tier-2 city coverage, making it an essential source for property records from cities like Tirunelveli, Erode, Vellore, Thanjavur, Tiruppur, Nagercoil, and Hosur — cities that are underrepresented on national-level real estate databases.

Housing.com is a technology-first real estate platform that provides particularly rich location and pricing data for urban properties, with strong coverage of Chennai, Coimbatore, and Madurai. Its data contributed to the price validation and locality-level classification in the dataset.

The Kaggle dataset was validated for internal consistency and cross-referenced against secondary sources to verify that price ranges, area ranges, and feature value distributions were reasonable for each city.

Government and Regulatory Sources

The second category of sources is official Tamil Nadu government databases used for data validation and verification purposes. The Tamil Nadu Registration Department, accessible through its TNREGINET online portal, maintains official records of all property registrations in Tamil Nadu including transaction values, property dimensions, and ownership details. This database was used to verify that the price ranges present in the assembled dataset were consistent with actual registered transaction values in each city, ensuring that the dataset does not contain systematically inflated or deflated asking prices relative to actual stamp duty valuations.

The Tamil Nadu Real Estate Regulatory Authority (TNRERA), established under the Real Estate (Regulation and Development) Act 2016, maintains a public registry of all approved residential projects in Tamil Nadu. This registry was used to validate the legitimacy of project-level listings in the dataset, particularly for newly constructed apartment and villa projects that form a significant portion of the data. The Chennai Metropolitan Development Authority (CMDA) and the respective urban local body planning authorities for other cities provided zoning and land use classification data that informed the locality categorization used in the dataset.

Geographic and Spatial Data Sources

The third category of sources is geographic data platforms used to compute the coordinate and proximity features in the dataset. The latitude and longitude coordinates of each property were obtained by geocoding the property addresses using the Google Maps Geocoding API, which converts textual addresses into precise GPS coordinate pairs. This geocoding process was applied to all 2,844 property records to ensure that every listing has accurate geographic coordinates for spatial distance computation.

The five proximity distance features — distance to city center, nearest school, nearest hospital, nearest metro station, and nearest highway — were computed using point-of-interest location data obtained from OpenStreetMap (OSM), the world's largest open-source geographic database maintained by a global community of contributors. For each property, the coordinates of the nearest instance of each amenity category were identified from the OSM database, and the distance between the property and the amenity was computed using the Haversine formula. The

Haversine formula calculates the great-circle distance between two points on the Earth's surface using their latitude and longitude coordinates, accounting for the curvature of the Earth and producing geodesic distances that are more accurate than simple Euclidean straight-line distances for geographic applications. The formula is expressed as:

$$d = 2R \times \arcsin(\sqrt{(\sin^2(\Delta lat/2) + \cos(lat_1) \times \cos(lat_2) \times \sin^2(\Delta lon/2))})$$

Where R is the Earth's radius (6,371 km), lat_1 and lon_1 are the coordinates of the property, and lat_2 and lon_2 are the coordinates of the nearest amenity. This formula was implemented in Python using the math module and applied to every property-amenity pair in the dataset, producing the five distance features that form the geospatial intelligence layer of PropAura's prediction model.

DATA CLEANING

5. DATA CLEANING

Data cleaning is one of the most critical steps in any machine learning project. Raw datasets collected from multiple real-world sources often contain duplicate records, missing values, and inconsistent formatting that can severely degrade model performance if left unaddressed. The Smart_real_estate.csv dataset containing 2,844 property records across 12 Tamil Nadu cities was subjected to a thorough three-stage data cleaning process before being used for model training. This section describes each stage in detail.

5.1 Remove Duplicates

In this project, the dataset Smart_real_estate.csv was collected by combining residential property listings from multiple real estate websites such as MagicBricks, 99acres, and Housing.com. Since property owners usually list the same property on multiple websites to reach more buyers, the same property sometimes appeared multiple times in the dataset. As a result, duplicate rows were present where all the property details such as city, locality, area, price, number of bedrooms, amenities score, and distance features were exactly the same.

When the dataset was analysed, it contained 2,844 total records. During duplicate checking, it was found that 608 rows were exact duplicate entries. These duplicate records did not provide any additional information and could negatively affect the performance of the machine learning model. If duplicates remain in the dataset, the model may learn the same property information multiple times and give higher importance to those repeated property features. This can reduce the model's ability to generalise well when predicting prices for new properties.

To remove duplicate records, the pandas drop_duplicates() function was used. This function removes repeated rows and keeps only the first occurrence of each unique property record. After applying this function, all duplicate entries were removed successfully.

After cleaning the dataset, the number of records reduced from 2,844 to 2,236 unique property listings. Each row in the dataset now represents a distinct residential property. The cleaned dataset was then used for further steps such as data analysis, model training, and evaluation.

5.2 Check Missing Values

After removing duplicate entries, the next step was to check the dataset for missing values. Missing values occur when certain information about a property is not recorded in the dataset. For example, a property listing might not include details such as area, number of bedrooms, or price. Missing data can affect machine learning models because algorithms require complete input values to learn patterns accurately.

To check for missing values, the isnull().sum() function in pandas was applied to all columns in the dataset. This function counts the number of null or empty values present in each column.

The result showed that there were no missing values in any of the 22 columns of the dataset. All 2,236 property records contained complete information for every feature such as city, property type, area, bedrooms, bathrooms, amenities score, distance features, and price.

Since the dataset contained no missing values, no data imputation or replacement techniques were required. The machine learning model was therefore trained using fully complete and reliable data, which helps improve the accuracy of the prediction system.

5.3 Correct Inconsistent Format

The third step of data preprocessing was to check the dataset for inconsistent formatting. Since the dataset was collected from multiple real estate platforms, there is a possibility that the same information might be written in different formats. Inconsistent formatting can create problems during machine learning because the model may treat similar values as different categories.

First, the categorical columns were checked. The city column contains properties from major Tamil Nadu cities such as Chennai, Coimbatore, Madurai, Trichy, Salem, Tirunelveli, Erode, Vellore, Thanjavur, Tiruppur, Hosur, and Nagercoil. All city names were written consistently using the same title-case format without spelling errors or abbreviations.

The property_type column was also verified to ensure that only the four categories — Apartment, Villa, Independent House, and Plot — were present. Similarly, the furnished column contained only three valid values: Furnished, Semi-Furnished, and Unfurnished. The facing column was checked to confirm that all eight directions such as North, South, East, West, North-East, North-West, South-East, and South-West were written in a consistent format. After performing these checks, it was confirmed that all columns were already properly formatted and consistent. Therefore, no corrections or format changes were required in this step.

SOFTWARE DESCRIPTION

6.SOFTWARE DESCRIPTION

Python

Python is the programming language used to build this project. It was chosen because it supports all the libraries needed for data science and machine learning. Every part of this project — data loading, cleaning, model training, prediction, visualization, and web application — was written in Python.

Google Colaboratory

Google Colab is a free cloud platform used to run this project. It was used to install libraries, upload the dataset, write and execute all the code, and launch the Streamlit web application through an ngrok public URL.

Streamlit (Version 1.41.0)

Streamlit is an open source Python library used to build interactive web applications for data science and machine learning projects without writing any HTML, CSS, or JavaScript. It converts Python scripts into fully functional web applications with interactive widgets, charts, and dynamic outputs that update in real time based on user input. Streamlit was installed in this project using the command pip install streamlit==1.41.0.

In this project, Streamlit was used to build the entire PropAura web application. The st.set_page_config function was used to set the page title as "PropAura — Tamil Nadu Market Intelligence", configure the layout to wide mode, and expand the sidebar by default. A sidebar navigation panel was built using st.radio, which allows the user to switch between the nine modules of the application — Smart Dashboard, Geo Intelligence, Price Prediction, Image Intelligence, Model Explainability, Investment Advisor, Property Comparator, EMI Calculator, and Blueprint Generator — by clicking on the module name in the sidebar.

The st.cache_data decorator was applied to the data loading function so that the dataset is read from the CSV file only once when the application first starts, and then served from memory for all subsequent requests without reading the file again. The st.cache_resource decorator was applied to the model training function so that the XGBoost model is trained only once per session and the trained model object is reused for every prediction request without retraining. This combination of caching made the application fast and responsive even when multiple users interact with it.

Throughout the nine modules, various Streamlit widgets were used to collect user inputs. st.slider was used to let users select numerical values like area in sqft, number of bedrooms, floor number, and age of the property. st.selectbox was used to let users choose categorical values like city, property type, and furnished status from dropdown menus. st.number_input was used to collect numeric inputs like loan amount, interest rate, and down payment in the EMI Calculator. st.button was used to trigger prediction and calculation actions. st.columns was used to arrange the layout into multiple side-by-side panels across all modules. st.plotly_chart was used to render all Plotly interactive charts. st.dataframe was used to display tabular data. st.download_button was used to provide the PDF download in the Blueprint Generator module. Custom HTML was injected using st.markdown with unsafe_allow_html=True to render KPI cards, styled section headers, badge labels, and score bars with custom colors and formatting that Streamlit's default widgets do not support.

Pandas

Pandas is a Python library used for data loading, manipulation, and analysis. It provides the DataFrame data structure, which organises data into rows and columns similar to a spreadsheet, and offers a wide range of functions to filter, group, sort, aggregate, and transform data. Pandas was installed as part of this project's dependencies.

In this project, Pandas was the first library used after the application starts. The pd.read_csv function was used to load the House_Price_prediction.csv file containing 2,844 residential property records from 12 Tamil Nadu cities into a pandas DataFrame. This DataFrame was used as the central data object throughout the entire application. During the data cleaning stage, df.duplicated().sum() was used to count duplicate rows and df.drop_duplicates() was used to remove them, reducing the dataset from 2,844 rows to 1,422 unique records. df.isnull().sum() was used to confirm that the dataset had zero missing values across all 22 columns.

During model training, df.copy() was used to create a working copy of the DataFrame so that the original data is preserved. The groupby function was used extensively in the Smart Dashboard module — df.groupby("year_listed")["price_inr"].median() was used to compute the median price for each year for the price trend chart, df.groupby("city")["price_inr"].median() was used to rank cities by median price for the city comparison bar chart, and df.groupby("furnished")["price_inr"].mean() was used to compare average prices across furnished, semi-furnished, and unfurnished properties. The df.sample(500) function was used to randomly select 500 properties for the scatter plot so the chart renders quickly without plotting all 2,844 points. The value_counts() function was used to compute property type distribution percentages for the pie chart.

NumPy

NumPy is a Python library for numerical computing. It provides support for fast mathematical operations on arrays and is used as the foundational numerical engine for most scientific Python libraries including Pandas, Scikit-learn, and XGBoost. NumPy was installed as part of this project's dependencies.

In this project, NumPy was used to handle the numerical outputs produced by the machine learning model and the SHAP explainer. When the XGBoost model's predict() function is called with a property's feature values, it returns the predicted price as a NumPy array. This array value was extracted and passed to the fmt_inr formatting function to display the price in lakhs or crores format. NumPy's np.clip function was used in the scoring modules to ensure that computed scores always stay within valid ranges — for example, keeping location quality scores between 0 and 10 even if the raw computation produces a value slightly outside this range due to edge cases in the data. In the Investment Advisor module, NumPy mathematical operations were used to compute compound annual growth projections over the selected investment horizon by applying exponential growth formulas to the base property price. NumPy was also used to handle array indexing when extracting individual SHAP values from the SHAP values matrix for display in the waterfall plot.

Scikit-learn

Scikit-learn is a Python machine learning library that provides a wide range of tools for data preprocessing, model selection, and model evaluation. It is built on top of NumPy and is one of the most widely used machine learning libraries in Python. Scikit-learn was installed as part of this project's dependencies.

In this project, three specific components of Scikit-learn were used. The first is LabelEncoder from sklearn.preprocessing. The dataset contains four categorical text columns — furnished, property_type, facing, and city — that cannot be directly used as inputs to the XGBoost model because the model works with numbers, not text. LabelEncoder was used to convert each of these columns into numeric values. A separate LabelEncoder object was fitted on each column individually. For example, the city column containing values like Chennai, Coimbatore, and Madurai was encoded into numbers like 0, 1, 2, and so on. The encoded values were added as new columns — furnished_enc, property_type_enc, facing_enc, and city_enc — to the training dataframe. A separate set of encoder objects was also stored and used during inference in the Price Prediction module, so that when a user selects "Chennai" from the dropdown, it is converted to the same numeric value that was used during training before being passed to the model.

The second component is train_test_split from sklearn.model_selection. This function was used to divide the dataset into a training set and a test set. The split was set to 80% training data and 20% test data using test_size=0.2. The random_state=42 parameter was set to ensure that the same split is produced every time the code runs, making the results fully reproducible. The training set was used to fit the XGBoost model, and the test set was kept completely separate and used only for evaluating the model's performance on data it has never seen.

The third component is mean_absolute_error and r2_score from sklearn.metrics. After the model was trained and predictions were generated on the test set, these two functions were used to measure prediction accuracy. mean_absolute_error computed the average absolute difference between the model's predicted prices and the actual prices in the test set, giving the MAE value displayed in the application sidebar. r2_score computed the R^2 coefficient of determination, which represents the proportion of price variance in the test set that the model successfully explains, also displayed in the sidebar to let users know the model's overall accuracy.

XGBoost (Version 1.7.6)

XGBoost, which stands for Extreme Gradient Boosting, is the core machine learning algorithm used in this project to predict residential property prices. It was installed using pip install xgboost==1.7.6. XGBoost is a gradient boosting algorithm that builds a large ensemble of decision trees one by one in a sequential manner. Each new tree that is added to the ensemble is specifically trained to fix the prediction errors that the previous combination of trees made. This process of sequential error correction, combined with built-in regularization that prevents overfitting, allows XGBoost to achieve very high prediction accuracy on structured tabular datasets like the property dataset used in this project.

In this project, the XGBRegressor class was used with the following configuration. n_estimators was set to 400, meaning the model builds 400 decision trees sequentially. learning_rate was set to 0.04, meaning each individual tree contributes only 4% of its prediction to the final result, which slows down the learning process and improves generalization by preventing any single tree from having too much influence. max_depth was set to 6, meaning each individual tree can have at most 6 levels of splits, allowing the model to capture complex relationships between features without becoming too deep and overfitting. subsample was set to 0.8, meaning each tree is trained on a randomly selected 80% of the training rows rather than all of them, which introduces randomness that reduces overfitting. colsample_bytree was set to 0.8, meaning each tree considers only a randomly

selected 80% of the 17 input features at each split, further increasing diversity across the ensemble. random_state was set to 42 to ensure all results are reproducible.

The model was trained on 17 input features: area_sqft, bedrooms, bathrooms, floor, total_floors, age_years, parking, amenities_score, furnished_enc, property_type_enc, facing_enc, city_enc, dist_city_center_km, dist_school_km, dist_hospital_km, dist_metro_km, and dist_highway_km. The target variable was price_inr. The model.fit() function was called on the 80% training split to train the model. After training, the model.predict() function is called every time a user enters property details in the Price Prediction module or the Property Comparator module, and the returned predicted price in rupees is formatted and displayed to the user.

SHAP (Version 0.46.0)

SHAP, which stands for SHapley Additive exPlanations, is a Python library used to explain the predictions made by machine learning models. It was installed using pip install shap==0.46.0. SHAP is based on Shapley values from cooperative game theory, which compute the fair contribution of each player in a game. Applied to machine learning, SHAP computes the contribution of each input feature to an individual prediction by calculating how much that feature pushed the predicted value above or below the average baseline prediction. For each property prediction, SHAP produces a value for every feature — a positive SHAP value means that feature increased the predicted price, and a negative SHAP value means it decreased the predicted price.

In this project, shap.TreeExplainer was used because it is specifically optimized for tree-based models like XGBoost and computes SHAP values much faster than the general explainer. The TreeExplainer object was created by passing the trained XGBoost model to it immediately after training, and was cached alongside the model using st.cache_resource. In the Model Explainability module, the explainer was used to compute SHAP values for all properties in the test set. These SHAP values were visualized using three types of SHAP plots. The waterfall plot shows for a single selected property how each of the 17 features contributed to pushing the predicted price up or down from the baseline average price to the final prediction. The bar plot shows the global average importance of each feature across all test properties by displaying the mean absolute SHAP value for each feature. The beeswarm plot shows the full distribution of SHAP values for every feature across all test samples, with each dot representing one property, allowing users to see not just which features matter most but how their impact varies across different types of properties. In the Price Prediction module, individual SHAP values were also computed for each user-submitted property to explain in rupees exactly which features made that specific property more or less expensive than the average.

Plotly

Plotly is a Python library used to create interactive data visualizations. Unlike static chart libraries, Plotly charts allow users to hover over data points to see exact values, zoom into specific regions, click on legend items to show or hide particular series, and pan across the chart — all directly in the browser without any additional setup. Plotly was installed as part of this project's dependencies.

In this project, two Plotly submodules were used. plotly.express, which provides high-level functions for quickly creating common chart types, and plotly.graph_objects, which provides lower-level control for building more customized visualizations. In the Smart Dashboard module, px.area was used to create the median price trend by year chart which shows how Tamil Nadu property prices changed from 2016 to 2023. px.bar was used to create the city-wise median price horizontal bar chart ranked from lowest to highest city median price, and also for the furnished status vs average price chart. px.scatter was used to create the area versus price scatter plot colored by city, which plots 500 randomly sampled properties to show the positive relationship between built-up area and price across all 12 cities. px.pie was used to create the property type distribution pie chart showing the percentage split between Apartments, Villas, Independent Houses, and Plots. In the Investment Advisor module, px.line was used to create the property value projection chart that shows how a property's value is estimated to grow over the user's selected investment horizon at different annual appreciation rate scenarios. In the EMI Calculator module, plotly.graph_objects was used to create a stacked bar chart showing the month-by-month amortization breakdown of each EMI payment into its principal repayment component and its interest payment component across the full loan tenure.

Folium (Version 0.16.0) and Streamlit-Folium (Version 0.20.0)

Folium is a Python library used to create interactive geographic maps. It works by generating Leaflet.js map objects in Python, which are then rendered as fully interactive maps in the browser where users can zoom, pan, click on markers, and switch between different map tile layers. Folium was installed using pip install folium==0.16.0. Streamlit-Folium is the companion library that allows Folium map objects to be embedded directly inside a Streamlit application, installed using pip install streamlit-folium==0.20.0.

In this project, Folium was used in both the Smart Dashboard module and the Geo Intelligence module to display interactive property maps. A folium.Map object was created centered on Tamil Nadu at latitude 10.8505 and longitude 78.6569 with an initial zoom level of 7 to show the entire state. Four different map tile layers were added — OpenStreetMap, Google Maps, Google Satellite, and Google Hybrid — and a

folium.LayerControl panel was added in the top right corner to let users switch between them. The Fullscreen plugin from folium.plugins was also added to allow the map to be expanded to full screen. For every property in the dataset, a folium.CircleMarker was placed at the property's GPS latitude and longitude coordinates. Each marker was colored according to its city using a predefined dictionary that assigns a unique color to each of the 12 Tamil Nadu cities. Clicking on any marker opens a popup that shows the property's city, price formatted in lakhs or crores, property type, area in sqft, BHK count, and furnished status. Custom HTML iframe markers were placed at the geographic center of each of the 12 cities showing the city name, average price, and total property count. In the Geo Intelligence module, additional markers were added to the map for nearby schools, hospitals, and banks retrieved from OpenStreetMap data for the user's selected locality. The st_folium function from streamlit-folium was used to embed the completed Folium map object into the Streamlit page and render it as a fully interactive map widget.

Pillow

Pillow is a Python image processing library that provides functions for opening, reading, and analyzing image files in various formats including JPEG, PNG, and WebP. It was installed as part of this project's dependencies.

In this project, Pillow was used in the Image Intelligence module. When a user uploads one or more property photographs using the st.file_uploader widget, the uploaded file is passed to Image.open() from the Pillow library to read the image data into memory. The image's pixel values are then accessed and analyzed to compute quality metrics including the average brightness of the image, the contrast level calculated from the spread of pixel intensity values, and the color saturation calculated from the image's color channel values. These computed metrics are displayed as visual score bars in the application and combined into an overall property quality score that gives the user a data-driven assessment of the property's condition and presentation quality based on its photographs.

ReportLab

ReportLab is a Python library used to create PDF documents programmatically by drawing text, shapes, lines, and tables onto a canvas object and saving the result as a PDF file. It was installed as part of this project's dependencies.

In this project, ReportLab was used in the Blueprint Generator module. After the user enters their desired property specifications — including the number of floors, number of rooms, plot dimensions, and preferred layout style — the application uses ReportLab to generate a complete PDF document. This PDF contains an engineering-style grid blueprint

drawing of the property floor plan with room dimensions and labels, a specifications summary table listing all the entered property details, the XGBoost model's predicted price for the specified configuration, and investment suitability information. The completed PDF is made available for download directly within the Streamlit interface using the st.download_button widget, so the user can save and share the blueprint report.

pyngrok

pyngrok is a Python library that wraps the ngrok tunneling tool, which creates a secure public HTTPS URL that forwards incoming internet traffic to a locally running server. It was installed as part of this project's dependencies using pip install pyngrok.

In this project, pyngrok was necessary because the PropAura Streamlit application runs on a localhost server inside Google Colab's virtual machine, which is not directly accessible from the public internet. When the Streamlit application is started on port 8501, pyngrok was used to create a public tunnel to that port, generating a publicly accessible HTTPS URL. This URL was printed in the Colab output and could be opened in any browser on any device to access the fully running PropAura application with all nine modules, real-time predictions, interactive charts, and maps — without requiring any separate cloud hosting or server configuration.

DESCRIPTIVE ANALYSIS

7.DESCRIPTIVE ANALYSIS

Descriptive analysis is an important stage in any data science project because it helps in understanding the overall structure and characteristics of the dataset. It summarizes the main features of the data and provides insights into the distribution and relationships between different variables. In this Smart Real Estate project, descriptive analysis was performed on the housing dataset to explore the important attributes related to residential properties and to understand how these attributes influence property prices. The dataset contains 2844 records with 22 different attributes that represent various aspects of real estate properties such as property size, number of rooms, furnishing status, location information, infrastructure accessibility, amenities, and price.

One of the major variables present in the dataset is the area of the property measured in square feet. The size of the property plays a crucial role in determining the value of a house. From the descriptive statistics, it can be observed that property areas vary significantly across the dataset, which indicates the presence of both small residential units and large luxury homes. Smaller properties are generally apartments designed for compact living, while larger properties represent villas or independent houses with more space and facilities. The variation in property size allows the dataset to represent different segments of the real estate market.

Another important attribute in the dataset is the number of bedrooms and bathrooms available in each property. These features are important indicators of the capacity and comfort level of a house. Most properties in the dataset have two or three bedrooms, which suggests that the houses are mainly designed for small to medium-sized families. Similarly, the number of bathrooms is closely related to the number of bedrooms and reflects modern housing standards where multiple bathrooms are provided for convenience. Houses with more bedrooms and bathrooms generally have higher market value because they offer greater comfort and usability for residents.

The dataset also contains information about the floor level and total number of floors in the building. These attributes are particularly relevant for apartment complexes and multi-storey buildings. Properties located on higher floors sometimes provide better ventilation, natural lighting, and scenic views, which may increase their desirability among buyers. On the other hand, some buyers prefer lower floors for easier accessibility. Therefore, floor information can have a noticeable influence on the pricing and demand for certain properties.

Another factor considered in the dataset is the age of the property, which represents how old the building is since construction. Property age can significantly impact the price and attractiveness of a house. Newly constructed properties are often preferred by buyers because they require less maintenance and provide modern design and infrastructure. In

contrast, older properties may have slightly lower prices because they may require renovation or repairs. Analyzing the age of properties helps in understanding how the real estate market values newly developed properties compared to older ones.

The dataset also includes the furnishing status of each property. Houses are categorized as furnished, semi-furnished, or unfurnished depending on the availability of furniture and interior fittings. Semi-furnished houses are commonly found in the dataset because they provide basic facilities such as kitchen cabinets, wardrobes, and lighting while allowing buyers to customize the rest of the interior according to their preferences. Furnished houses usually have higher prices since they are ready for immediate occupancy and reduce the additional cost of furnishing for buyers.

Location is one of the most important factors in real estate valuation, and the dataset includes several attributes that describe the geographical and infrastructural aspects of each property. These include the distance from the city center, schools, hospitals, metro stations, and highways. Properties located closer to the city center typically have higher demand because they provide easier access to workplaces, shopping areas, and other important facilities. Similarly, houses located near schools and hospitals are preferred by families because they ensure convenience and safety.

The dataset also considers transportation accessibility, which is represented by the distance to metro stations and highways. Good connectivity to transportation networks improves the convenience of commuting and increases the overall value of residential properties. Properties located near highways or metro stations allow residents to travel easily to different parts of the city. Therefore, transportation infrastructure is a significant factor influencing housing prices.

Another unique feature included in the dataset is the amenities score, which represents the number and quality of facilities available within the residential complex. These amenities may include security services, parking spaces, parks, recreational areas, gyms, and other lifestyle facilities. Properties with higher amenities scores generally provide a better living experience and are therefore more attractive to potential buyers. As a result, such properties tend to have higher market prices compared to those with fewer facilities.

The dataset also includes geographical coordinates such as latitude and longitude, which represent the exact location of each property. These coordinates are useful for spatial analysis and visualization, as they allow properties to be mapped geographically. Location-based analysis can help identify property clusters, high-demand areas, and price variations across different regions.

Finally, the dataset contains the property price, which is the most important variable in this project. Property prices vary widely depending on multiple factors such as property size, number of rooms, location advantages, amenities, and accessibility to infrastructure. Some properties in the dataset belong to the affordable housing category, while others represent premium or luxury residential units. The variation in prices reflects the diversity of the housing market and provides a good foundation for building predictive models.

Overall, the descriptive analysis provides a clear understanding of the structure and characteristics of the real estate dataset. It highlights how different property attributes contribute to variations in housing prices and helps identify important variables that can be used for further analysis. These insights are useful for preparing the dataset for data visualization, correlation analysis, and machine learning model development. By understanding the patterns and relationships present in the data, the Smart Real Estate system can provide more accurate predictions and valuable insights for buyers, sellers, and real estate investors.

SYSTEM DESIGN

8. SYSTEM DESIGN

System Design is the process of defining the architecture, components, and structure of a system to ensure it functions efficiently. It describes how different modules of the system interact with each other to process data and achieve the desired output.

8.1 IDENTIFY RELATIONSHIP

System design is an important stage in the development of any software or data science application. It focuses on defining the structure of the system, the interaction between different components, and the relationships between various data elements. In the Smart Real Estate Price Prediction system, identifying relationships between different variables is essential because property price is influenced by several factors such as location, property size, infrastructure availability, amenities, and housing characteristics. Understanding these relationships helps in building a structured system that can analyze property data efficiently and produce accurate predictions.

The dataset used in this project contains multiple attributes related to residential properties, including area, number of bedrooms, number of bathrooms, property age, furnishing status, floor details, amenities score, and distances to important facilities such as schools, hospitals, metro stations, and highways. Each of these attributes is related either directly or indirectly to the final property price. Identifying these relationships helps in understanding how changes in one variable may affect another variable within the system. For example, the size of the property is directly related to the price of the house. Larger houses generally have higher prices because they provide more space and better living conditions.

Another important relationship in the system exists between the number of bedrooms, bathrooms, and the property area. These variables are closely connected because larger houses usually contain more rooms and facilities. A property with more bedrooms and bathrooms tends to be more comfortable and suitable for larger families, which increases its demand in the housing market. Therefore, these attributes collectively influence the price of the property and play a significant role in the prediction process.

Location-related attributes also have strong relationships within the system. The dataset includes several location-based factors such as distance to the city center, distance to schools, distance to hospitals, and distance to transportation infrastructure like metro

stations and highways. Properties located closer to the city center or important facilities usually have higher market value because they offer better accessibility and convenience. These relationships help the system understand how location advantages contribute to price variations across different properties.

The dataset also includes an amenities score that represents the availability of facilities within a residential complex. This score is related to the quality of living provided by the property. Residential projects with better amenities such as parks, security services, parking spaces, and recreational facilities are more attractive to buyers. As a result, properties with higher amenities scores generally have higher prices. Identifying this relationship helps the system incorporate lifestyle factors when predicting property values.

Another relationship exists between the age of the property and its market price. Newly constructed properties often have higher demand because they provide modern designs, improved infrastructure, and lower maintenance requirements. Older properties may experience a decrease in value due to wear and tear or outdated facilities. By identifying the relationship between property age and price, the system can better understand how depreciation affects real estate values.

Furnishing status is another attribute that influences property pricing. Properties that are fully furnished typically have higher prices compared to semi-furnished or unfurnished houses because they include additional interior features and ready-to-use facilities. This relationship allows the system to consider the level of interior readiness when estimating property values.

In addition to these relationships, the dataset also includes geographical coordinates such as latitude and longitude, which represent the exact location of each property. These coordinates are related to location-based analysis and can help identify clusters of high-value or low-value properties. Such spatial relationships are useful for understanding how different areas influence housing prices.

Overall, identifying relationships between these variables is essential for designing an effective Smart Real Estate system. It helps in understanding how different property attributes interact with each other and influence the final property price. These relationships form the foundation for further analysis, including correlation analysis and machine learning model development. By clearly defining these connections, the system can process real estate data more efficiently and generate accurate predictions for property valuation.

DATA IMPLEMENTATION

9. DATA IMPLEMENTATION

Data Implementation refers to the stage where the collected dataset is processed and implemented using programming to generate results. In this project, it includes writing the source code, creating visualizations to understand the data, and evaluating the model accuracy for property price prediction.

9.1 SOURCE CODE

CELL:1

```
import warnings, logging, os
warnings.filterwarnings("ignore")
os.environ["PYTHONWARNINGS"] = "ignore"
logging.getLogger().setLevel(logging.ERROR)
!pip install -q --no-warn-conflicts \
  streamlit==1.41.0 \
  pyngrok \
  folium==0.16.0 \
  streamlit-folium==0.20.0 \
  plotly \
  xgboost==1.7.6 \
  shap==0.46.0 \
  scikit-learn \
  pandas \
  numpy \
  pillow \
  reportlab 2>/dev/null
import logging
logging.getLogger().setLevel(logging.ERROR)
print("✅ All done! Run Cell 2 now 👉")
```

CELL:2

```
import pandas as pd
df = pd.read_csv("Smart_real_estate.csv")
```

```
print("Duplicates:", df.duplicated().sum())
df.drop_duplicates(inplace=True)
print("Duplicates after cleaning:", df.duplicated().sum())
```

CELL:3

```
%%writefile app.py
import os
os.environ["STREAMLIT_SERVER_ENABLE_STATIC_SERVING"] = "true"
os.environ["STREAMLIT_BROWSER_GATHER_USAGE_STATS"]   = "false"
import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import folium
from streamlit_folium import st_folium
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.preprocessing import LabelEncoder
import xgboost as xgb
import shap
from PIL import Image
import math, time, warnings, os
warnings.filterwarnings("ignore")
html, body, .stApp,
div[data-testid="stAppViewContainer"],
div[data-testid="stHeader"],
section[data-testid="stSidebar"],
.main, .block-container {
   background-color: #ffffff !important;
   color: #1a1a1a !important;}
```

```
p, span, div, label, h1, h2, h3, h4, h5, h6,
li, td, th, strong, em, small,
[data-testid="stWidgetLabel"] p,
[data-testid="stWidgetLabel"],
[data-baseweb="menu"] li:hover {
   background-color: #e8f0fe !important;}
.stButton > button:hover {
   transform: translateY(-2px) !important;
   box-shadow: 0 6px 20px rgba(26,115,232,.4) !important;}
.kpi {
   margin-bottom: 20px;}
.b-blue   { background:#e8f0fe; color:#1a73e8 !important; border:1px solid #1a73e8; }
.b-green  { background:#e6f4ea; color:#2e7d32 !important; border:1px solid #2e7d32; }
.b-red    { background:#fce8e6; color:#c62828 !important; border:1px solid #c62828; }
.b-gold   { background:#fef9e7; color:#f57f17 !important; border:1px solid #f57f17; }
.b-purple { background:#f3e8fd; color:#6a1b9a !important; border:1px solid #6a1b9a; }
""", unsafe_allow_html=True)
     <div style="display:flex;justify-content:space-between;margin-bottom:3px">
       <span style="font-size:.8rem;color:#8b949e">{label}</span>
       <span style="font-size:.8rem;font-
weight:600;color:{color}">{score:.1f}/{maxv}</span> </div>
     <div class="sb-wrap"><div class="sb-fill" style="width:{pct}%;background:linear-
gradient(90deg,{color},{color}88)"></div></div>
   </div>""", unsafe_allow_html=True)
def haversine(lat1, lon1, lat2, lon2):
   R = 6371
dlat = math.radians(lat2-lat1); dlon = math.radians(lon2-lon1)
   a = math.sin(dlat/2)**2 +
math.cos(math.radians(lat1))*math.cos(math.radians(lat2))*math.sin(dlon/2)**2
@st.cache_data
def load_data():
```

```
    csv = "/content/Smart_real_estate.csv"
    if not os.path.exists(csv):
        st.error("❌ House_Price_prediction.csv not found! Upload it alongside this app.")
        st.stop()
    return pd.read_csv(csv)
df = load_data()
@st.cache_resource
def train_model(data):
    d = data.copy()
    for col in ["furnished","property_type","facing","city"]:
        le = LabelEncoder()
        d[col+"_enc"] = le.fit_transform(d[col])
       features = [
        "area_sqft","bedrooms","bathrooms","floor","total_floors","age_years",
        "parking","amenities_score","furnished_enc","property_type_enc",
        "facing_enc","city_enc",
        "dist_city_center_km","dist_school_km","dist_hospital_km",
        "dist_metro_km","dist_highway_km"]
    X = d[features]; y = d["price_inr"]
    Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.2, random_state=42)
    m = xgb.XGBRegressor(n_estimators=400, learning_rate=0.04, max_depth=6,
                         subsample=0.8, colsample_bytree=0.8,
                         random_state=42, verbosity=0)
    m.fit(Xtr, ytr)
    </div><hr>""", unsafe_allow_html=True)
        page = st.radio("", [
        "🏠 Smart Dashboard","📍 Geo Intelligence", "🏡 Price Prediction",
        "🖼️ Image Intelligence","🧠 Model Explainability", "📈 Investment Advisor",
        "🔁 Property Comparator","💸 EMI Calculator", "📐 Blueprint Generator",
    ], label_visibility="collapsed")
```

```
    st.markdown("<hr>", unsafe_allow_html=True)
    st.markdown(f"""
    <div style="font-size:.72rem;color:#8b949e;text-align:center;line-height:1.8">
      Dataset: <b style="color:#58a6ff">{len(df):,} properties</b><br>
      Cities: Chennai · CBE · Madurai · Salem · Tirunelveli · Erode · Vellore · Thanjavur · Tiruppur · Hosur · Nagercoil · Trichy<br>
      Model R²: <b style="color:#3fb950">{r2v:.3f}</b>  |  MAE: <b>{fmt_inr(mae)}</b><br><br>
<span style="color:#3fb950">● System Online</span>
    </div>""", unsafe_allow_html=True)
if page == "🏠 Smart Dashboard":
    sec_header("🏠", "Smart Dashboard", "Real-time Tamil Nadu property market overview")
    avg_p   = df["price_inr"].mean()
    avg_psf = (df["price_inr"] / df["area_sqft"]).mean()
    top_loc = df.groupby("locality")["price_inr"].median().idxmax()
    top_city= df.groupby("city")["price_inr"].median().idxmax()
    st.markdown("<br>", unsafe_allow_html=True)
    cl, cr = st.columns([3,2])
    with cl:
        st.markdown("#### 📈 Price Trend by Year Listed")
        trend = df.groupby("year_listed")["price_inr"].median().reset_index()
        fig = px.area(trend, x="year_listed", y="price_inr",
                      labels={"price_inr":"Median Price (₹)","year_listed":"Year"},
                      color_discrete_sequence=["#58a6ff"])
        fig.update_traces(fillcolor="rgba(88,166,255,.12)", line_color="#58a6ff")
        fig.update_layout(**PLOTLY_LAYOUT,
            xaxis=dict(showgrid=False,color="#8b949e"),
            yaxis=dict(gridcolor="#30363d",color="#8b949e"))
        st.plotly_chart(fig, use_container_width=True)
    with cr:
```

```
st.markdown("#### 🗺️ Median Price by City")
city_med =
df.groupby("city")["price_inr"].median().sort_values(ascending=True).reset_index()
fig2 = px.bar(city_med, x="price_inr", y="city", orientation="h",
              color="price_inr", color_continuous_scale="Blues",
              labels={"price_inr":"Median Price","city":""})
fig2.update_layout(**PLOTLY_LAYOUT, coloraxis_showscale=False,
   yaxis=dict(color="#8b949e",tickfont=dict(size=11)),
   xaxis=dict(gridcolor="#30363d",color="#8b949e"))
st.plotly_chart(fig2, use_container_width=True)
ca, cb = st.columns([2,3])                f"{fmt_inr(int(df['price_inr'].min()))} –
{fmt_inr(int(df['price_inr'].max()))}",
            "Min to Max"),
            unsafe_allow_html=True)
folium.LayerControl(position="topright").add_to(m)
   CITY_COLORS_MAP = {
   "Chennai":"#1a73e8", "Coimbatore":  "#2e7d32","Madurai":     "#f57f17",
   "Trichy":"#c62828", "Salem":"#6a1b9a", "Tirunelveli": "#00838f",
   "Erode":"#e65100", "Vellore":"#1565c0", "Thanjavur":   "#4527a0",
   "Tiruppur":"#558b2f", "Hosur":"#d84315","Nagercoil":   "#00695c",}
CITY_CENTERS_MAP = {
   "Chennai":     (13.0827, 80.2707),"Coimbatore":  (11.0168, 76.9558),
   "Madurai":     (9.9252,  78.1198),"Trichy":      (10.7905, 78.7047),
   "Salem":       (11.6643, 78.1460),"Tirunelveli": (8.7139,  77.7567),
   "Erode":       (11.3410, 77.7172),"Vellore":     (12.9165, 79.1325),
   "Thanjavur":   (10.7870, 79.1378),"Tiruppur":    (11.1085, 77.3411),
   "Hosur":       (12.7409, 77.8253),"Nagercoil":   (8.1833,  77.4119),}
sample = df.sample(min(400, len(df)))
for _, row in sample.iterrows():
            {fmt_inr(row['price_inr'])} </b></td></tr>
        </table> </div>""", max_width=220)).add_to(m)
```

```
    for city, (clat, clon) in CITY_CENTERS_MAP.items():
        clr = CITY_COLORS_MAP[city]
        city_data = df[df["city"] == city]
        avg_p     = fmt_inr(int(city_data["price_inr"].mean()))
        count     = len(city_data)
        folium.Marker([clat, clon],popup=folium.Popup(f"""
            <div style='font-family:Arial;min-width:160px;padding:4px'>
              <div style='font-weight:700;font-size:14px;
                        color:{clr};margin-bottom:6px'>🗺️ {city} </div>
              <table style='font-size:12px;color:#333'>
                <tr><td>📊 Listings</td>
                    <td><b>{count}</b></td></tr> <tr><td>💰 Avg Price</td>
                    <td><b style='color:#1a73e8'> {avg_p}
                    </b></td></tr> <tr><td>📍 Localities</td>
            <td><b>{city_data['locality'].nunique()}</b> </td></tr> </table
        🗺️ PropAura City Map
      </div><div style='margin-bottom:6px;color:#555;font-size:10px'>
with cd:
    st.markdown("#### 🏘️ Property Type Distribution")
    pt = df["property_type"].value_counts().reset_index()
    pt.columns = ["type","count"]
    fig4 = px.pie(pt, names="type", values="count",
                  color_discrete_sequence=["#58a6ff","#3fb950","#e3b341","#a371f7"],
                  hole=0.45)
    fig4.update_layout(**PLOTLY_LAYOUT, showlegend=True,
        legend=dict(bgcolor="#1c2128",bordercolor="#30363d"))
    st.plotly_chart(fig4, use_container_width=True)
 with ce:
    st.markdown("#### 🛋️ Avg Price by Furnishing")
    furn = df.groupby("furnished")["price_inr"].mean().reset_index()
```

```
        fig5 = px.bar(furn, x="furnished", y="price_inr",
                      color="furnished",
                      color_discrete_sequence=["#f78166","#e3b341","#3fb950"],
                      labels={"price_inr":"Avg Price (₹)","furnished":""})
        fig5.update_layout(**PLOTLY_LAYOUT, showlegend=False,
            xaxis=dict(color="#8b949e"),
            yaxis=dict(gridcolor="#30363d",color="#8b949e"))
        st.plotly_chart(fig5, use_container_width=True)
elif page == "📍 Geo Intelligence":
    sec_header("📍", "Geo Intelligence",
               "Real location analytics — schools, hospitals, banks & more via
OpenStreetMap")
    with col_in:
        st.markdown("#### 📍 Select Location")
        city_sel    = st.selectbox("City", sorted(df["city"].unique()), key="geo_city")
        all_locs    = sorted(df[df["city"] == city_sel]["locality"].unique())
        loc_sel     = st.selectbox("Locality", all_locs, key="geo_loc")
        use_manual  = st.checkbox("Override with manual coordinates")
        if use_manual:
            mlat = st.number_input("Latitude",value=CITY_CENTERS[city_sel][0],
                                   format="%.5f")
        radius   = st.slider("Search Radius (km)", 1, 10, 3)
        analyze  = st.button("🔍 Analyse Location")
                lat = row["latitude"].mean()
                lon = row["longitude"].mean()
            cc  = CITY_CENTERS[city_sel]
            dc  = haversine(lat, lon, cc[0], cc[1])
            sub = df[df["locality"] == loc_sel]
            TYPE_MAP = {
                "🏫 Schools":     ("amenity", "school",      "#1a73e8", "🏫"),
```

```
        "🏥 Hospitals":    ("amenity", "hospital",    "#c62828", "🏥"),
        "🏦 Banks":        ("amenity", "bank",        "#f57f17", "🏦"),
        "🛒 Supermarkets": ("shop",    "supermarket", "#2e7d32", "🛒"),
    grade      = ("A+" if overall >= 8.5 else
                  "A"  if overall >= 7.0 else
                  "B+" if overall >= 6.0 else
                  "B"  if overall >= 5.0 else "C")
    grade_color = ("#3fb950" if overall >= 7 else
                   "#e3b341" if overall >= 5 else "#f78166")
    loc_avg  = sub["price_inr"].mean() if len(sub) > 0 else 0
    city_avg = df[df["city"] == city_sel]["price_inr"].mean()
    vs_city  = ((loc_avg - city_avg) / city_avg * 100
                if city_avg > 0 else 0)
    cc,
        popup=f"{r['locality']}<br>{fmt_inr(r['price_inr'])}
    legend_html = ""<div style='position:fixed;bottom:20px;left:20px;
    z-index:9999;background:white;border:2px solid #ccc;border-radius:10px;
    padding:10px 14px;font-size:12px; box-shadow:0 4px 12px rgba(0,0,0,0.15)'
    else:
    st.info("No named facilities found in this area."
            " Try increasing the search radius.")
elif page == "🏠 Price Prediction":
  sec_header("🏠", "Smart Price Prediction", "XGBoost AI valuation with SHAP explainability")
  col_f, col_o = st.columns([1, 1.45])
    facing    = st.selectbox("Facing Direction", ["East","North-East","North","North-West","West","South","South-West","South-East"])
    amenities = st.slider("Amenities Score (0-10)", 0, 10, 6)
    st.markdown("#### 📍 Distances")
    dm = st.slider("Distance to Metro/Rail (km)", 0.1, 8.0, 1.5, 0.1)
```

```
        ds = st.slider("Distance to School (km)", 0.1, 5.0, 0.8, 0.1)
        dh = st.slider("Distance to Hospital (km)", 0.1, 5.0, 1.2, 0.1)
        dw = st.slider("Distance to Highway (km)", 0.1, 8.0, 2.0, 0.1)
        predict_btn = st.button("🚀 Predict Price")
    with col_o:
            conf = round(min(96, max(72, r2v * 100 - abs(np.random.normal(0,1.5)))), 1)
            loc_avg = df[df["locality"]==locality]["price_inr"].mean() if len(loc_row)>2
else pred
            <span class="badge b-blue">Low: {fmt_inr(low)}</span>
            <span class="badge b-green">High: {fmt_inr(high)}</span>
            <span class="badge b-purple">{prop_type}</span> </div>
          </div>""", unsafe_allow_html=True)
          c1,c2,c3,c4 = st.columns(4)
          c1.markdown(kpi_card("Price/sqft", f"₹{ppsf:,.0f}"), unsafe_allow_html=True)
          c2.markdown(kpi_card("Confidence", f"{conf}%", color="var(--green)"),
unsafe_allow_html=True)
          c3.markdown(kpi_card("Investment", inv_rat), unsafe_allow_html=True)
          est_rent = int(pred * 0.003)
          c4.markdown(kpi_card("Est. Monthly Rent", fmt_inr(est_rent)),
unsafe_allow_html=True)
elif page == "🖼️ Image Intelligence":
    sec_header("🖼️", "Image Intelligence", "Upload multiple room photos — AI predicts
property quality & price range")
    st.markdown("""
    <div class="res-card" style="margin-bottom:18px">
      <b style="color:#1a73e8">📸 How it works:</b>
      <div style="font-size:.85rem;color:#555555;margin-top:6px;line-height:1.9">
        Upload photos of each room → AI analyses brightness, contrast, texture, luxury
score per room
        → Combines all room scores → Predicts overall property grade + estimated price
    """, unsafe_allow_html=True)
    ROOMS = [
```

```
("🏠 Exterior / Front View", "exterior"),("🛋️ Living Room","living"),
("🛏️ Master Bedroom","bedroom"),("🍳 Kitchen","kitchen"),
("🚿 Bathroom","bathroom"),("🏡 Other Room / Balcony","other"), ]
st.markdown("#### 📤 Upload Room Photos (upload as many as you have)
img_city = st.selectbox("City", sorted(df["city"].unique()), key="img_city"
        score_bar("🌞 Brightness & Lighting", res["brightness"])
        score_bar("🎨 Contrast & Clarity",   res["contrast"],   color="#3fb950")
        score_bar("🧱 Texture Richness",     res["texture"],    color="#e3b341")
        score_bar("🌈 Color Harmony",        res["color_harm"], color="#f78166")
        score_bar("📐 Space Perception",     res["space"],      color="#a371f7")
st.markdown("---") st.markdown("### 🏠 Overall Property Assessment")
overall_score = round(sum(room_scores) / len(room_scores), 1)
overall_grade = ("A+" if overall_score>=8.5 else "A" if overall_score>=7.5 else
            "B+" if overall_score>=6.5 else "B" if overall_score>=5.5 else "C")
st.markdown("#### 📊 Room Score Comparison")
fig_rooms = go.Figure(go.Bar(
  x=list(room_results.keys()),
  y=[r["overall"] for r in room_results.values()],
  marker_color=["#3fb950" if r["overall"]>=7 else "#e3b341" if r["overall"]>=5
            else "#f78166" for r in room_results.values()],
  text=[f"{r['overall']}/10" for r in room_results.values()],
  textposition="outside" ))
fig_rooms.add_hline(y=overall_score, line_dash="dash", line_color="#1a73e8",
            annotation_text=f"Avg: {overall_score}", annotation_position="right")
st.plotly_chart(fig_rooms, use_container_width=True)
cats  = ["Brightness","Contrast","Texture","Color","Space"]
verdict = ("🏆 Excellent property — premium finishes, great lighting and space
across all rooms."
        if overall_score >= 8 else
```

```
            "✅ Good quality property — well maintained with decent interiors."
            if overall_score >= 6.5 else
            "🔶 Average property — some rooms need improvement."
            if overall_score >= 5 else
            "⚠️ Below average — significant renovation recommended before listing.")
        reno = ("No renovation needed" if overall_score>=8 else
                "Minor touch-ups"      if overall_score>=6.5 else
                "Moderate renovation"  if overall_score>=5 else
                "Full renovation recommended"
        """, unsafe_allow_html=True)
elif page == "🧠 Model Explainability":
    sec_header("🧠", "Model Explainability", "SHAP-based Explainable AI — why prices are what they are")
    c1,c2 = st.columns([1,2])
    bullets = ""
    for _, r in top3.iterrows():
        sign  = "increased" if r["SHAP"]>0 else "decreased"
        color = "#3fb950"   if r["SHAP"]>0 else "#f78166"
        bullets += f'<li><span style="color:{color};font-weight:600">{r["Feature"]}</span> {sign} price by <b>{fmt_inr(abs(r["SHAP"]))}</b></li>'
    st.markdown(f"""
    <div class="res-card">
    </div>""", unsafe_allow_html=True)
    st.markdown("#### Actual vs Predicted Prices (Test Set)")
    act = y_test.values[:150]
    prd = model.predict(X_test[:150])
    fig_avp = go.Figure(
    st.plotly_chart(fig_avp, use_container_width=True)
elif page == "📈 Investment Advisor":
```

```
    sec_header("📈", "Investment Advisor", "AI-powered Tamil Nadu investment
recommendations")
    col_f, col_r = st.columns([1,1.6])
    with col_f:
        horizon   = st.selectbox("Investment Horizon", ["Short (1-2 yrs)","Medium (3-5
yrs)","Long (7+ yrs)"])
        risk      = st.radio("Risk Appetite", ["Conservative","Moderate","Aggressive"])
        go_btn    = st.button("🎯 Get Recommendations")
    with col_r:
        if go_btn:
            with st.spinner("Analyzing Tamil Nadu market…"):
                time.sleep(1.2)
                growth_map = {"Short (1-2 yrs)":6,"Medium (3-5 yrs)":11,"Long (7+ yrs)":14}
                filt["apprec"]   = filt["roi_sc"] * 1.3 + growth_map.get(horizon,11) * 0.5
                ).sort_values("ROI",ascending=False).head(6).reset_index()
            st.markdown("#### 🏆 Top Recommended Areas")
            medals = ["🥇","🥈","🥉","4️⃣","5️⃣","6️⃣"]
            for rank,(_, r) in enumerate(summary.iterrows(), 1):
                st.markdown(f"""
elif page == "🔁 Property Comparator":
        city = st.selectbox(f"City_{key}", sorted(df["city"].unique()),
                            index=list(sorted(df["city"].unique())).index(def_city) if def_city in
df["city"].unique() else 0,
                            key=f"city_{key}", label_visibility="collapsed")
        v2 = prop_inputs("🏠 Property B","p2","Coimbatore",2,900,2,10,0)
    if st.button("⚖️ Compare Now"):
        def predict_v(v):
            row = df[df["locality"]==v["loc"]]
            inp = np.array([[v["area"],v["bhk"],v["bat"],v["fl"],v["fl"]+5,
        with cb:
            wb2 = winner_badge(p2,p1,v2["area"],v1["area"])
```

```
cats = ["Area","BHK","Bathrooms","Floor","Amenities","Value/sqft"]
def norm(v,mn,mx): return round((v-mn)/(mx-mn)*10,1) if mx>mn else 5.0
    fillcolor="rgba(63,185,80,.14)", line_color="#3fb950"))
fig_rad.update_layout(
polar=dict(radialaxis=dict(visible=True,range=[0,10],color="#8b949e",gridcolor="#3036
3d"),
bank_rates = {"SBI Home Loan (8.50%)": 8.50,"HDFC Bank (8.70%)":8.70,
"ICICI Bank (8.75%)": 8.75,"Axis Bank (8.75%)":8.75,"LIC Housing (8.65%)":9.65,
    "Bank of Baroda (8.40%)": 8.40,"Custom Rate":None,}
    fig_pie = px.pie(
        names=["Principal Amount", "Total Interest"],
        values=[int(loan_amt), int(total_int)],hole=0.45,
        color_discrete_sequence=["#1a73e8", "#f78166"])
    fig_pie.update_layout(**PLOTLY_LAYOUT, height=280,
        legend=dict(bgcolor="#ffffff", bordercolor="#dddddd",
                font=dict(color="#1a1a1a")))
    st.plotly_chart(fig_pie, use_container_width=True)
    balance  = loan_amt
    yearly   = {}
    for month in range(1, n + 1):
        cmp_rows.append({
            "Bank":          bk.split(" (")[0],
            "Interest Rate":  f"{rt}%",
            "Monthly EMI":    fmt_inr(int(em)),
            "Total Interest": fmt_inr(int(tot - loan_amt)),
            "Total Payment":  fmt_inr(int(tot)),
else:
    st.markdown("""
    <div class="res-card" style="text-align:center;padding:80px">
      <div style="font-size:3rem">💸</div>
      <div style="font-size:1.15rem;font-weight:600;color:#1a73e8;margin:14px 0">
```

```
            Home Loan EMI Calculator </div>
           <div style="font-size:.85rem;color:#8b949e">
            Enter your property price and loan details<br>
            on the left, then click Calculate EMI </div>
         </div>""", unsafe_allow_html=True)
elif page == " 📐 Blueprint Generator":
   sec_header(" 📐 ", "Blueprint Generator",
             "Engineering grid blueprint · Furniture · PDF Download · Side-by-side
Compare")
   import matplotlib.pyplot as plt
   import matplotlib.patches as patches
   from matplotlib.patches import FancyBboxPatch
   from matplotlib.lines import Line2D
   import io
   ROOM_COLORS = {"Living Room":"#C8E6FA","Master Bedroom": "#C8F0D8",
      "Bedroom 2":"#FFE0B2","Bedroom 3":"#FFCDD2","Bedroom 4":"#E1BEE7",
      "Bedroom 5":"#B2EBF2","Master Bath":"#D1C4E9","Bathroom 2":#CFD8DC",
      "Bathroom 3":"#CFD8DC","Kitchen":"#FFF9C4","Dining":"#DCEDC8",
      "Balcony":"#B2DFDB","Study/Pooja":"#FFE0B2","Garage":"#ECEFF1",}
INTERIOR_HEX = {
      "Living Room":
{"Walls":"#F5F0E8","Ceiling":"#FFFFFF","Floor":"#D4A96A","Accent":"#2C5F8A"},
      "Master Bedroom":
{"Walls":"#E8F4F0","Ceiling":"#FFFFFF","Floor":"#8B6914","Accent":"#1A6B3C"},
      "Bedroom 2":
{"Walls":"#FDF5E6","Ceiling":"#FFFFFF","Floor":"#A0522D","Accent":"#E67E22"},
      "Bedroom 3":
{"Walls":"#FDE8E8","Ceiling":"#FFFFFF","Floor":"#8B4513","Accent":"#C0392B"},
      "Bedroom 4":
{"Walls":"#EAF6FF","Ceiling":"#FFFFFF","Floor":"#7B5E3A","Accent":"#1A3A5C"},
      "Bedroom 5":
{"Walls":"#F3E8FD","Ceiling":"#FFFFFF","Floor":"#8B7355","Accent":"#6A1B9A"},
```

```
    "Kitchen":
{"Walls":"#FFFDE7","Ceiling":"#FFFFFF","Floor":"#607D8B","Accent":"#F39C12"},
    "Master Bath":
{"Walls":"#EDE7F6","Ceiling":"#FFFFFF","Floor":"#455A64","Accent":"#7B1FA2"},
    "Garage":
{"Walls":"#ECEFF1","Ceiling":"#B0BEC5","Floor":"#546E7A","Accent":"#37474F"},}
  FURNITURE = {
    "Living Room":   [("Sofa",0.35,0.15,"#8B7355"),("Coffee
Table",0.45,0.38,"#A0522D"),
                ("TV Unit",0.35,0.75,"#4A4A4A"),("Chair",0.72,0.25,"#8B7355")],
    "Master Bedroom":
[("Bed",0.30,0.25,"#8B6914"),("Wardrobe",0.75,0.15,"#5C4827"),
                ("Dresser",0.75,0.65,"#7B5E3A"),("Side Table",0.55,0.20,"#A0896B")],
    "Kitchen":      [("Counter",0.15,0.50,"#B0BEC5"),("Stove",0.15,0.20,"#607D8B"),
                ("Sink",0.15,0.75,"#90A4AE"),("Fridge",0.75,0.15,"#CFD8DC")],
    "Master Bath":   [("Toilet",0.20,0.20,"#ECEFF1"),("Sink",0.75,0.20,"#ECEFF1"),
                ("Shower",0.55,0.65,"#B2EBF2")],
    "Bathroom 2":    [("Toilet",0.20,0.20,"#ECEFF1"),("Sink",0.75,0.20,"#ECEFF1")],
    "Dining":       [("Table",0.40,0.40,"#8B6914"),("Chair",0.20,0.30,"#6D4C41"),
                ("Chair",0.20,0.55,"#6D4C41"),("Chair",0.65,0.30,"#6D4C41"),
                ("Chair",0.65,0.55,"#6D4C41")],
    "Balcony":      [("Plant",0.20,0.20,"#2E7D32"),("Chair",0.60,0.50,"#795548")],
    "Study/Pooja":   [("Desk",0.25,0.40,"#8D6E63"),("Chair",0.50,0.55,"#6D4C41"),
                ("Shelf",0.75,0.25,"#5D4037")]  }
  def calc_rooms(area, bedrooms, bathrooms, hall, kitchen,
            dining, balcony, study, garage):
    ratios = {}
    if hall:   ratios["Living Room"]  = 0.18
    for i in range(bedrooms):
      Line2D([0],[0], color="#2196F3", lw=2.5,
          ls="--", label="Window"),
      patches.Patch(facecolor="white",
```

```
                edgecolor="#1A3A5C",
                label="Door arc"),
        patches.Patch(facecolor="#E8F0F8",
                hatch="///",
                edgecolor="#555",
                label="Wet area", alpha=0.5),]
            color="#1A3A5C", lw=2)
    ax.text(74, 7, "Scale: 3 m",
            ha="center", fontsize=9,
            color="#1A3A5C", fontweight="bold")
    plt.tight_layout(pad=0.5)
    return fig
def fig_to_bytes(fig, dpi=180):
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=dpi,
                bbox_inches="tight")
    buf.seek(0)
    return buf.read()
    story  = []
    for img_bytes, label in zip(img_bytes_list, labels):
        story.append(Paragraph(label, styles["Heading2"]))
        story.append(Spacer(1, 0.3*cm))
        img_buf = io.BytesIO(img_bytes)
        story.append(RLImage(img_buf,
                             width=25*cm, height=18*cm))
        story.append(Spacer(1, 0.5*cm))
    doc.build(story)
    buf.seek(0)
    return buf.read()
        st.image(buf, use_container_width=True)<div style="text-
align:center;color:#8b949e;font-size:.73rem;padding:8px 0">
```

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Cities: Chennai · Coimbatore · Madurai · Trichy · Salem + 6 more |

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"""", unsafe_allow_html=True)

CELL:4

```
with open("app.py", "r") as f:
    code = f.read()
old = '''def kpi_card(label, val, sub="", color="#1a73e8"):
sub_html = f\'<div style="font-size:.72rem;color:#2e7d32;margin-top:5px">{sub}</div>\' if sub else ""
        f\'letter-spacing:.9px;margin-bottom:6px">{label}</div>\'
with open("app.py", "w") as f:
    f.write(code)
print("✅ KPI fix done!")
```

CELL:5

```
with open("app.py", "r") as f:
code = code.replace( 'paper_bgcolor="#1c2128",font_color="#e6edf3"',
    'paper_bgcolor="#ffffff",font_color="#1a1a1a"')
with open("app.py", "w") as f:
    f.write(code)
print("✅ Charts fix done!")
```

CELL:6

```
with open("app.py", "r") as f:
    code = f.read()
code = code.replace('tiles="CartoDB dark_matter"','tiles="OpenStreetMap"')
code = code.replace("tiles=\"CartoDB dark_matter\"","tiles=\"OpenStreetMap\"")
code = code.replace('mapbox_style="carto-darkmatter"','mapbox_style="open-street-map"')
code = code.replace('st_folium(m, width=700, height=380)','st_folium(m, width=700, height=380, returned_objects=[])')
```

```
with open("app.py", "w") as f:
    f.write(code)
print("✅ Maps fix done!")
```

CELL:7

```
with open("app.py", "r") as f:
    code = f.read()
code = code.replace(
    '''[data-baseweb="tag"] {background-color: #1a73e8 !important;}
[data-baseweb="tag"] span { color: #ffffff !important;}''',
    '''[data-baseweb="tag"] {background-color: #1a73e8 !important;
border-radius: 4px !important;}
[data-baseweb="tag"] span,[data-baseweb="tag"] div,[data-baseweb="tag"] * {
    color: #ffffff !important;font-weight: 600 !important;}''')
with open("app.py", "w") as f:
    f.write(code)
print("✅ Multiselect fix done!")
```

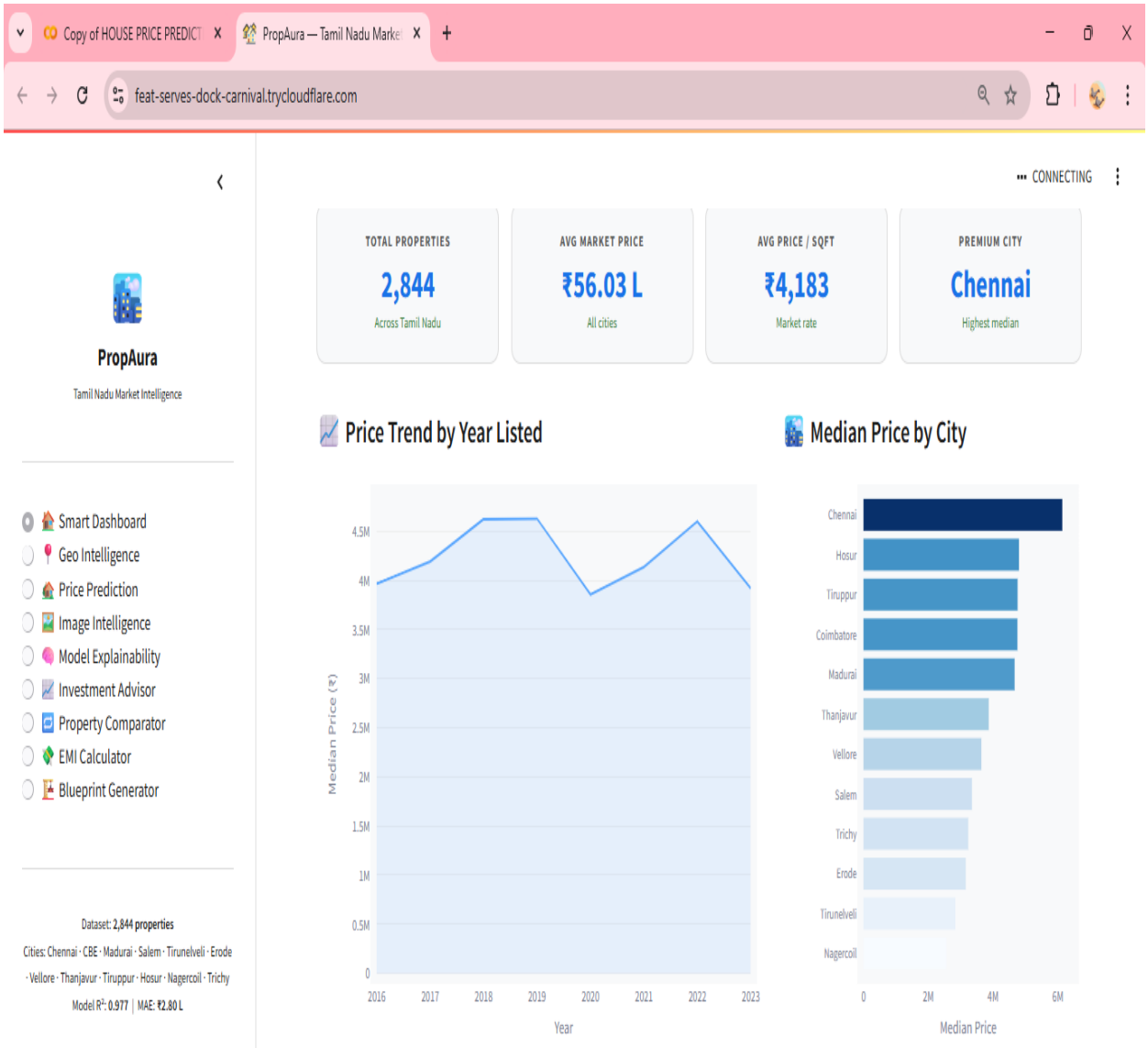
CELL:8

```
import subprocess, time, re, os
os.system("pkill -f streamlit 2>/dev/null")
os.system("pkill -f cloudflared 2>/dev/null")
time.sleep(3)
os.system("wget -q https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64 -O cloudflared")
print("⏳ Starting Streamlit... waiting 8 seconds") time.sleep(8)
tunnel_proc = subprocess.Popen( ["./cloudflared", "tunnel", "--url", "http://localhost:8501"], stdout=subprocess.PIPE, stderr=subprocess.PIPE)
print("⏳ Starting tunnel... waiting 12 seconds") time.sleep(12)
for _ in range(30):
    line = tunnel_proc.stderr.readline().decode("utf-8")
    if not line:
```

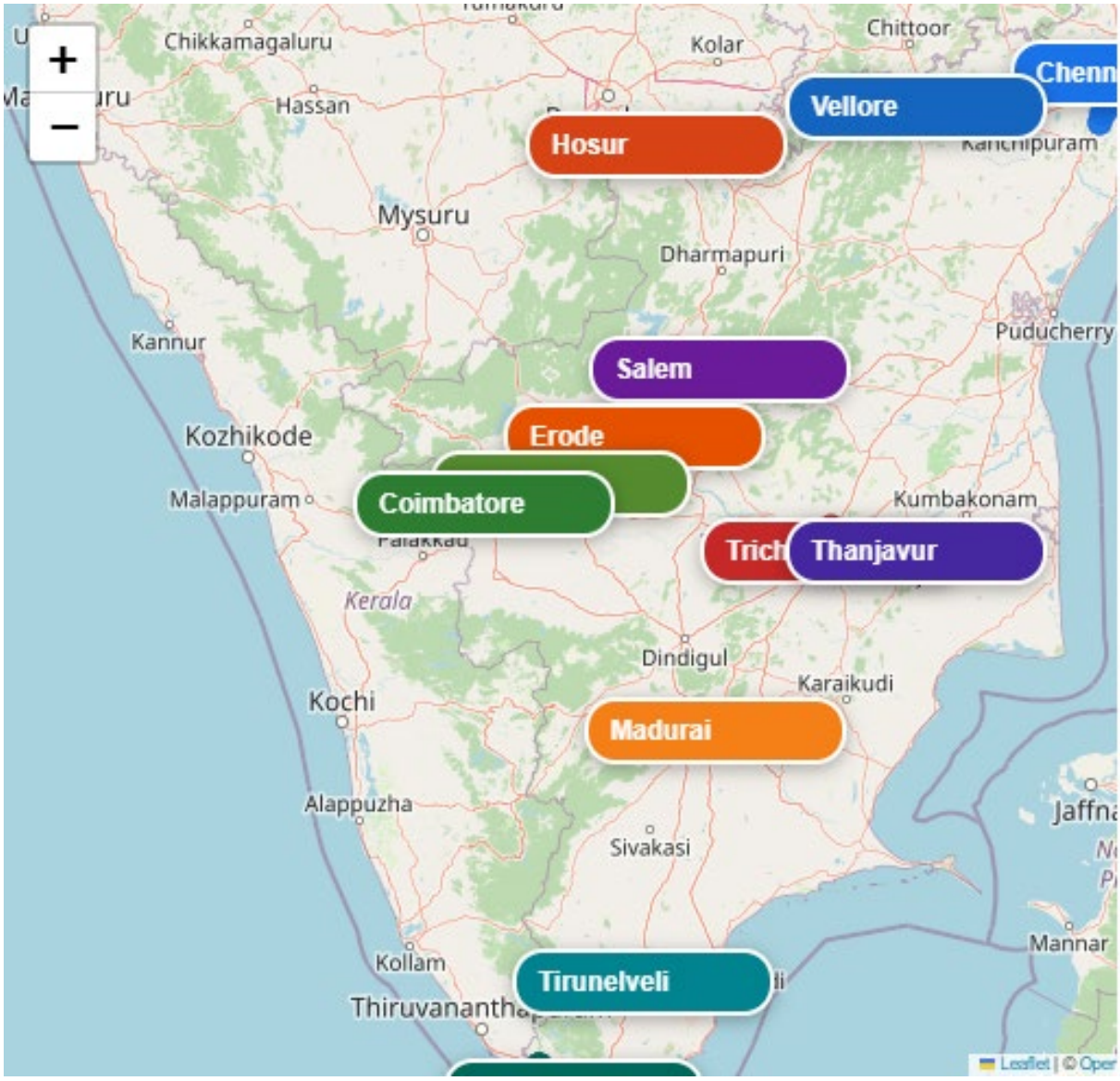
```
        break
    match = re.search(r"https://[a-zA-Z0-9\-]+\.trycloudflare\.com", line)
    if match:
        url = match.group(0)
        print("=" * 60)
        print("🚀 Propaura AI is LIVE!")
        print(f"🌐 URL: {url}")
        print("=" * 60)
        break
else:
    print("⚠️ URL not found. Try running this cell again."
```

9.2 VISUALIZATION

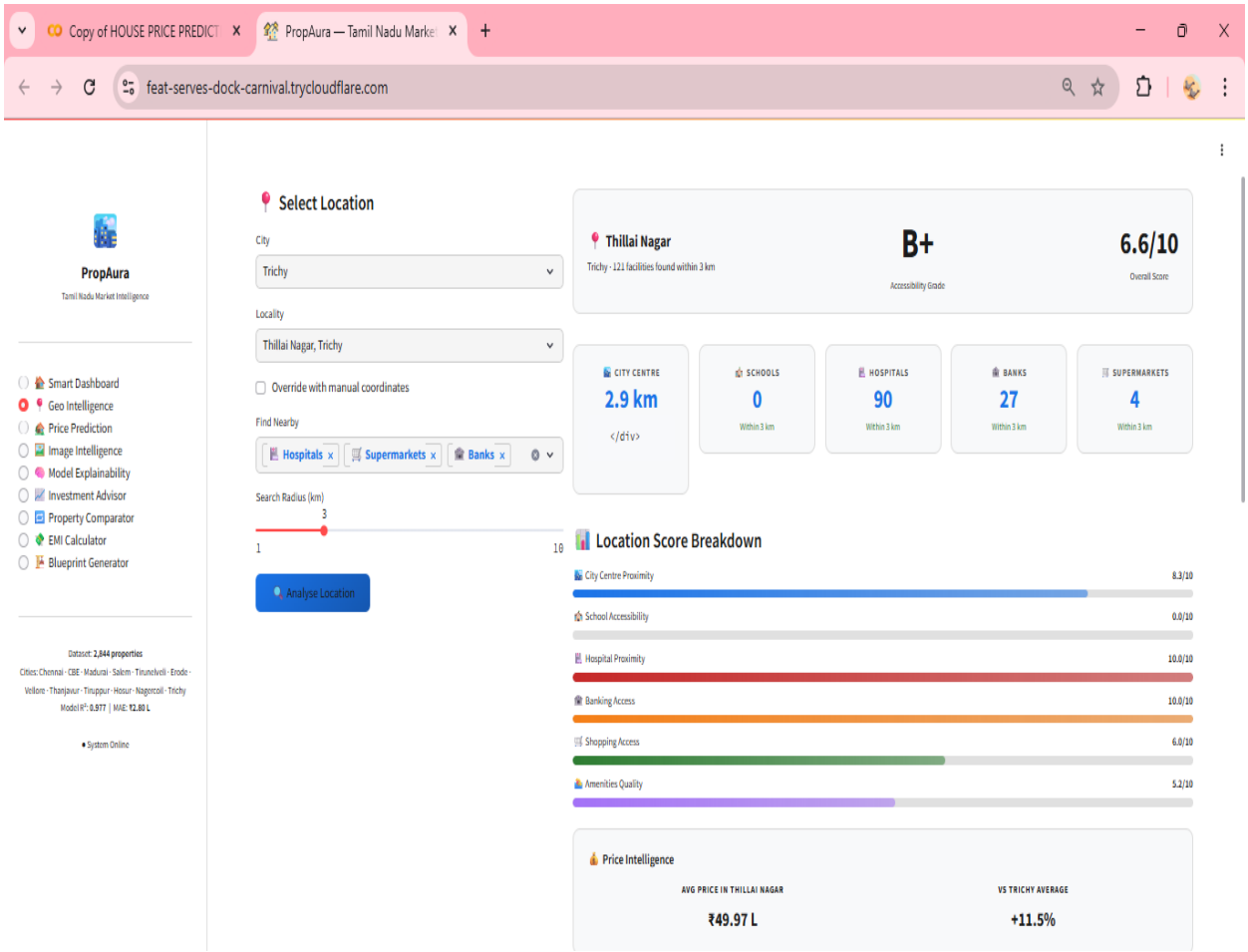
SMART DASHBOARD:



LOCATION MAP:



GEO INTELLIGENCE:



SMART PRICE PREDICTION:

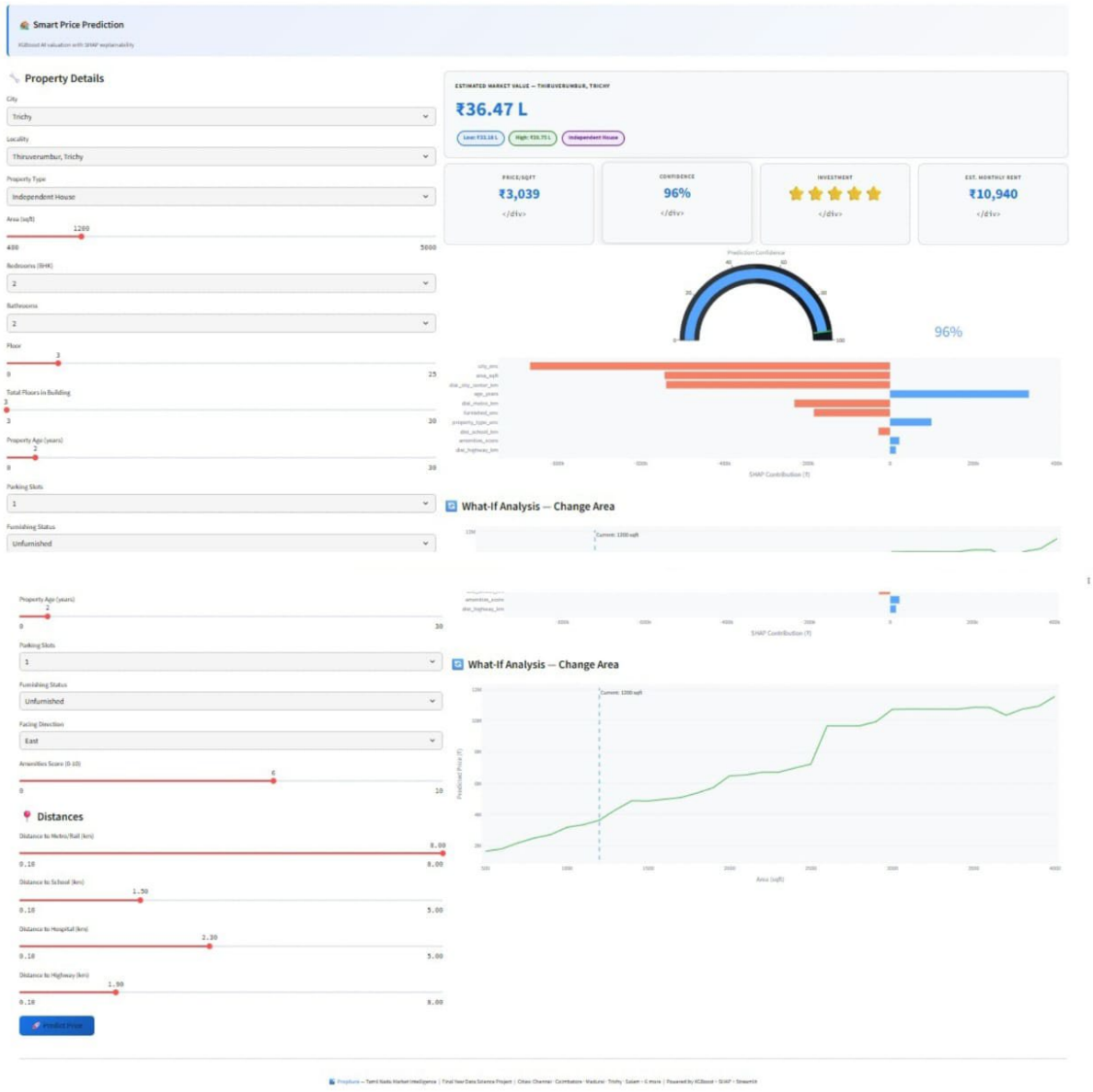
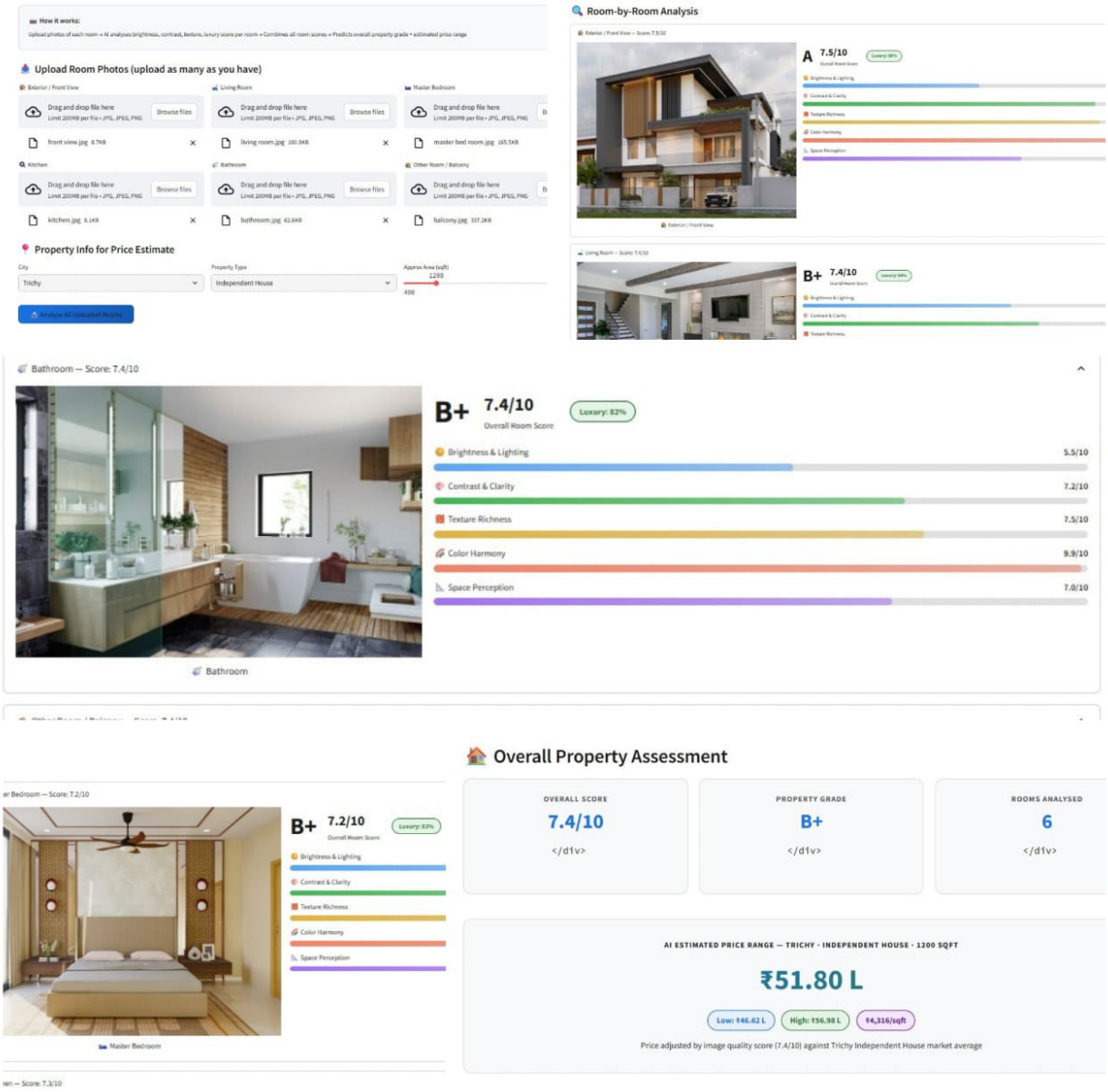
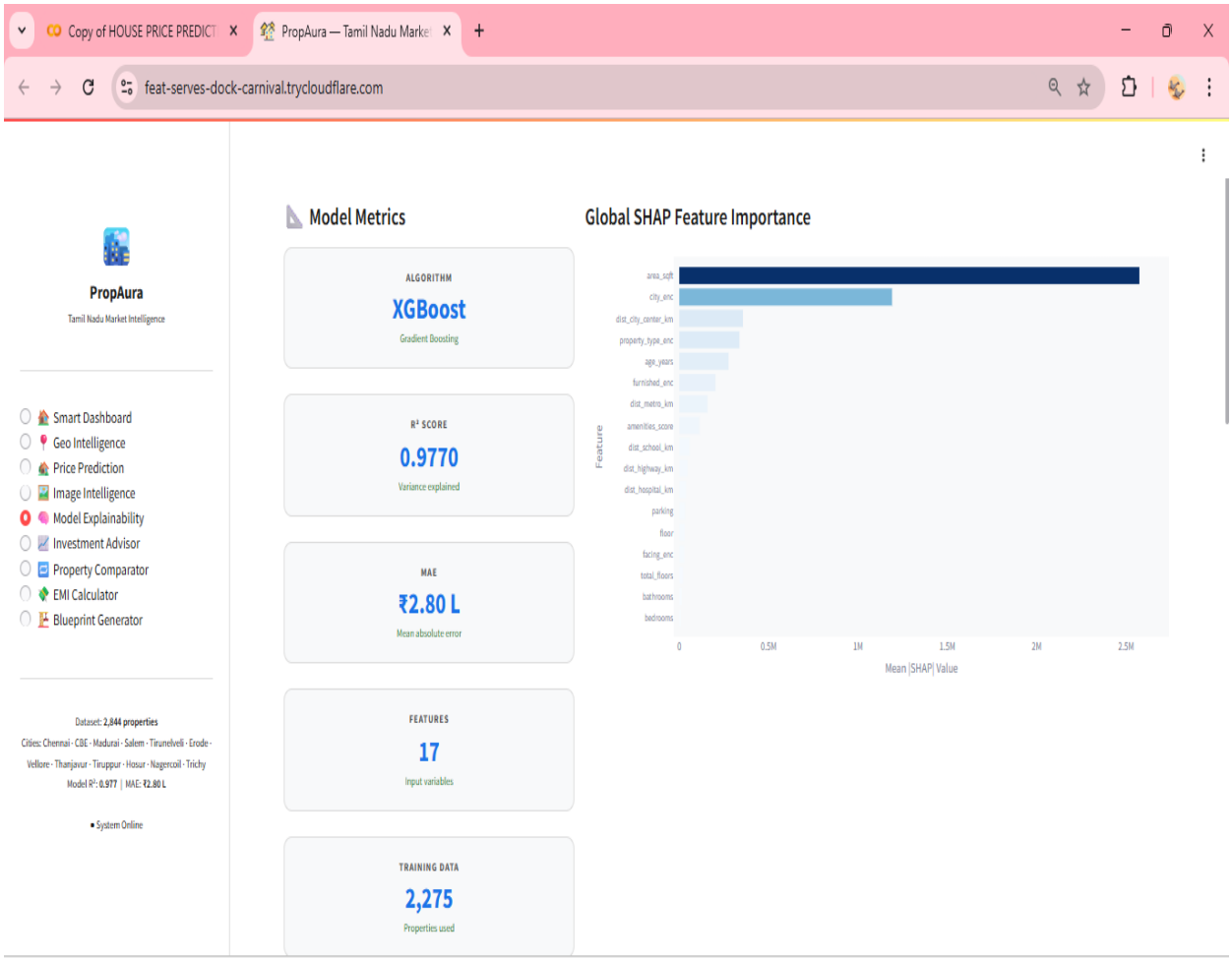


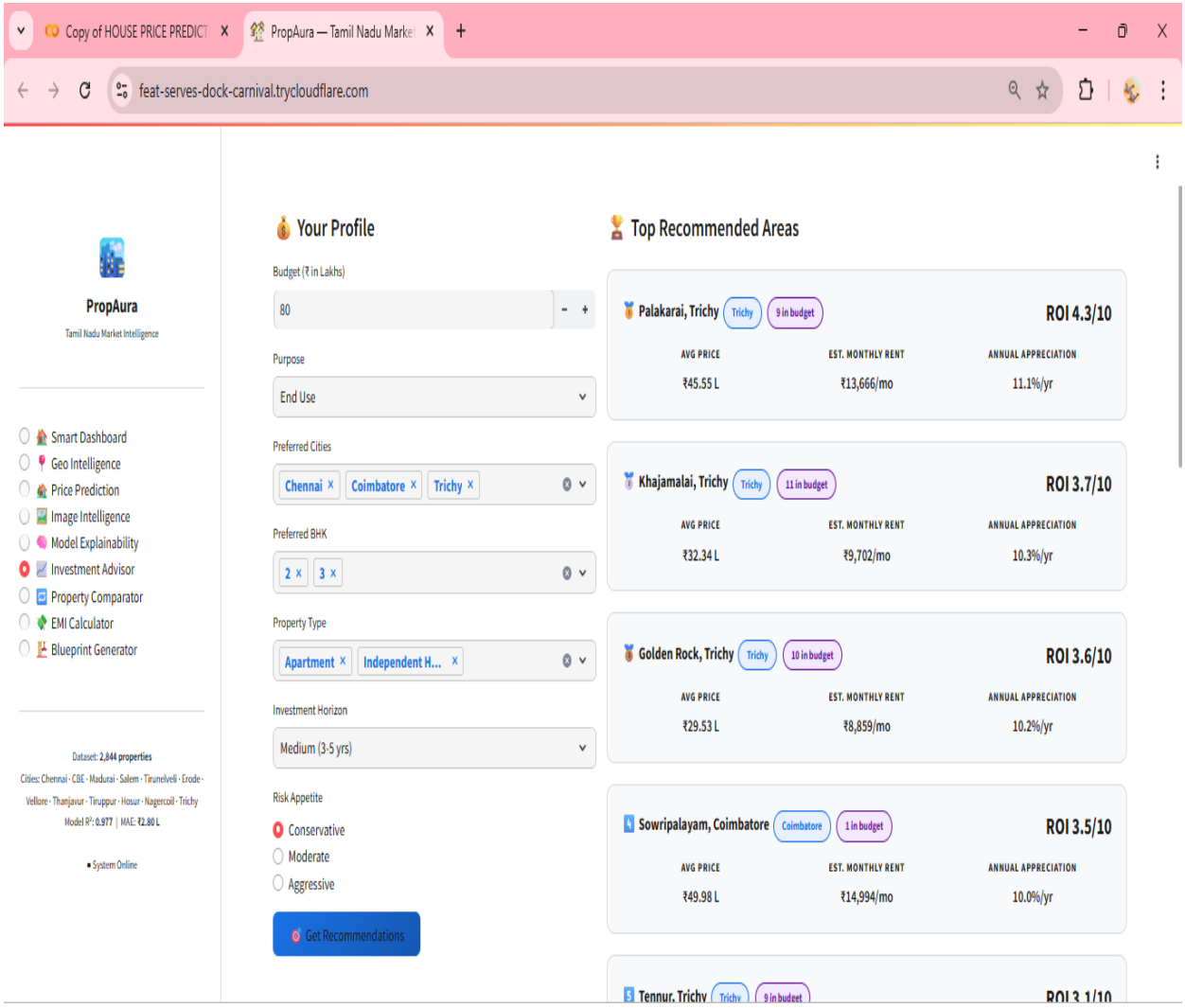
IMAGE INTELLIGENCE:



MODEL EXPLAINABILITY:



INVESTMENT ADVISOR:



LOAN BREAKDOWN:

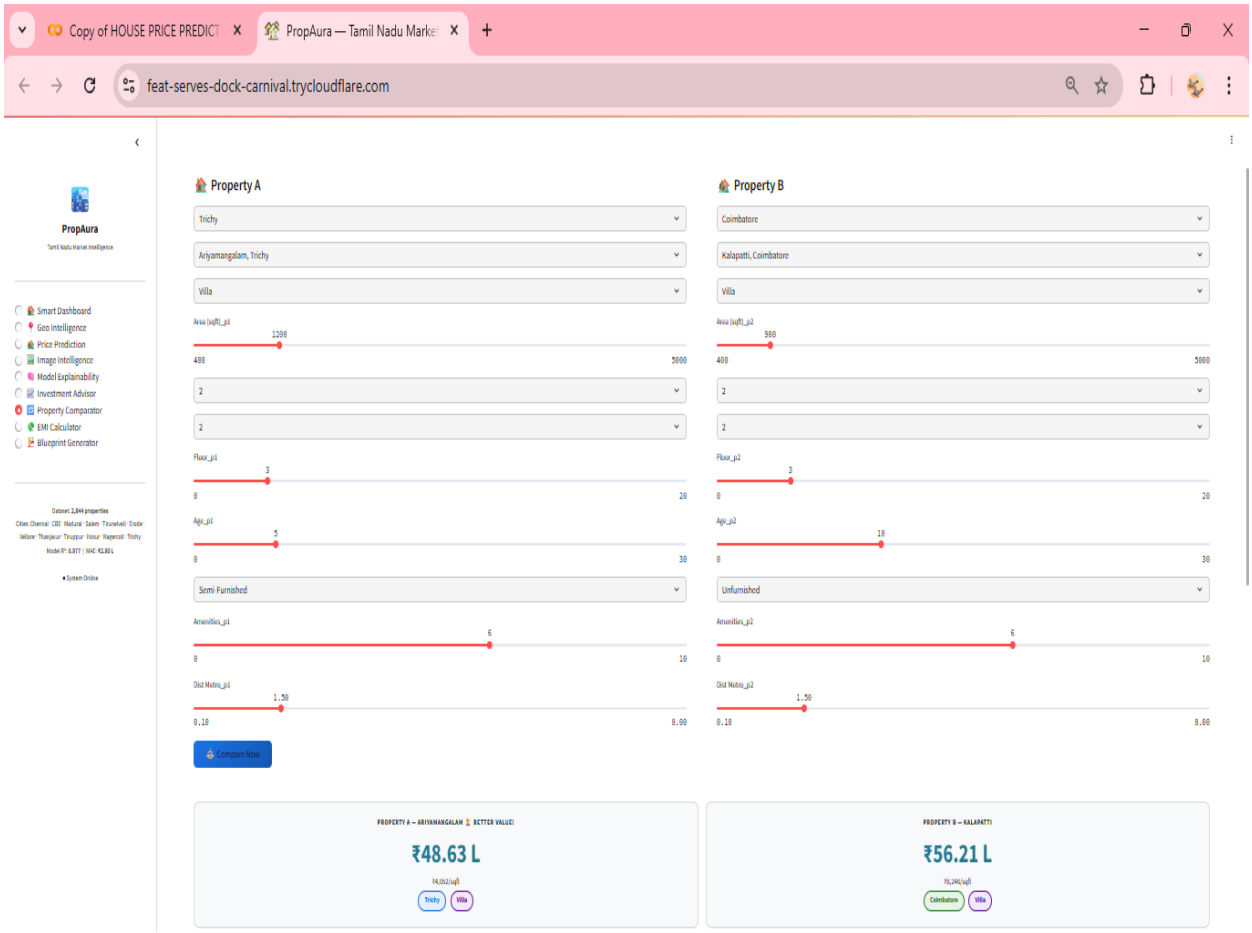
📋 Year-by-Year Breakdown Table

Year	Principal Paid	Interest Paid	Balance Left
1	₹1.17 L	₹5.14 L	₹58.83 L
2	₹1.28 L	₹5.04 L	₹57.55 L
3	₹1.39 L	₹4.92 L	₹56.16 L
4	₹1.52 L	₹4.80 L	₹54.64 L
5	₹1.66 L	₹4.66 L	₹52.98 L
6	₹1.80 L	₹4.51 L	₹51.18 L
7	₹1.97 L	₹4.35 L	₹49.21 L
8	₹2.14 L	₹4.17 L	₹47.07 L
9	₹2.34 L	₹3.98 L	₹44.73 L
10	₹2.55 L	₹3.77 L	₹42.18 L

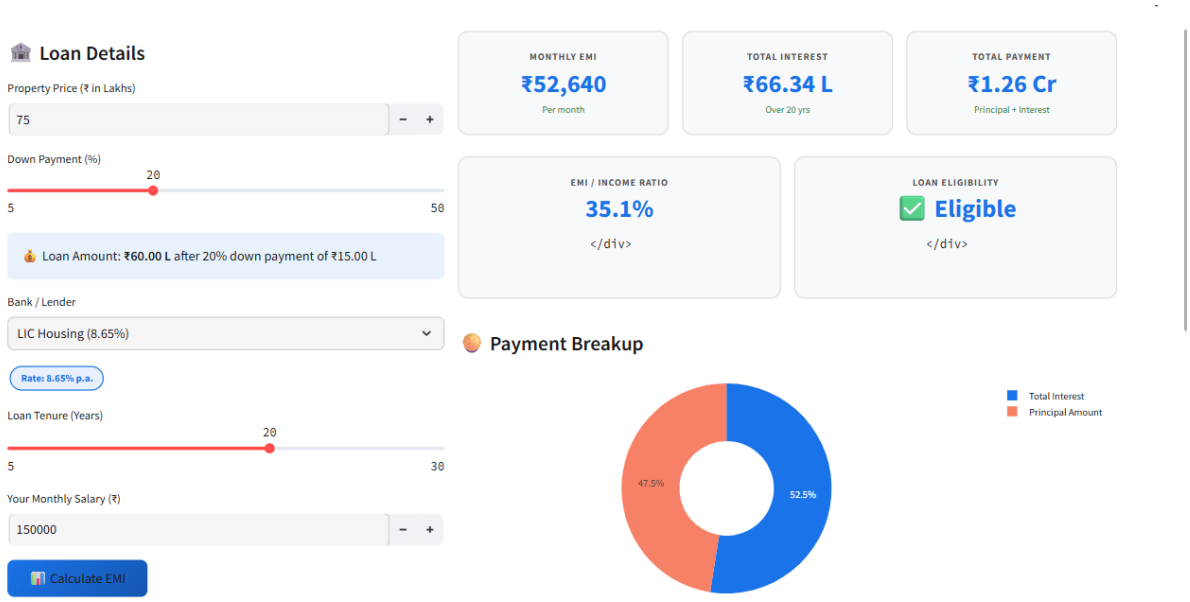
🏦 Compare All Banks for Your Loan

Bank	Interest Rate	Monthly EMI	Total Interest	Total Payment	Savings vs Max
SBI Home Loan	8.5%	₹52,069	₹64.97 L	₹1.25 Cr	—
HDFC Bank	8.7%	₹52,831	₹66.80 L	₹1.27 Cr	₹45,904
ICICI Bank	8.75%	₹53,022	₹67.25 L	₹1.27 Cr	₹0
Axis Bank	8.75%	₹53,022	₹67.25 L	₹1.27 Cr	₹0
LIC Housing	8.65%	₹52,640	₹66.34 L	₹1.26 Cr	₹91,734
Bank of Baroda	8.4%	₹51,690	₹64.06 L	₹1.24 Cr	₹3.20 L

PROPERTY COMPARATOR:



EMI CALCULATOR:



BLUEPRINT GENERATOR:

Room Area Breakdown

Room	Area (sqft)	% of Total	Dimensions
Living Room	225	18.8%	≈ 13.5 × 16.5 ft
Master Bedroom	175	14.6%	≈ 11.9 × 14.6 ft
Bedroom 2	139	11.5%	≈ 10.6 × 12.9 ft
Bedroom 3	138	11.5%	≈ 10.6 × 12.9 ft
Master Bath	75	6.2%	≈ 7.8 × 9.5 ft
Bathroom 2	75	6.2%	≈ 7.8 × 9.5 ft
Bathroom 3	75	6.2%	≈ 7.8 × 9.5 ft
Kitchen	125	10.4%	≈ 10.1 × 12.3 ft
Dining	112	9.3%	≈ 9.5 × 11.6 ft
Balcony	62	5.2%	≈ 7.1 × 8.7 ft

Interior Design Color Suggestions

Living Room
Walls: Ivory White | Ceiling: Pure White | Floor: Golden Oak Wood
Accent: Ocean Blue

Master Bedroom
Walls: Mint Cream | Ceiling: Pure White | Floor: Walnut Brown
Accent: Forest Green

Bedroom 2
Walls: Linen White | Ceiling: Pure White
Accent: Burnt Orange

Bedroom 3
Walls: Blush Pink | Ceiling: Pure White | Floor: Mahogany
Accent: Deep Red

Master Bath
Walls: Lilac White | Ceiling: Pure White | Floor: Charcoal Tile
Accent: Deep Purple

Bathroom 2
Walls: Cool Grey | Ceiling: Pure White
Accent: Steel Blue

Vastu Analysis

Vastu Score

North / East facing — excellent for natural light and positive energy.

- Master Bedroom → **South-West** corner (stability & rest)
- Kitchen → **South-East** corner (fire element)
- Pooja / Study → **North-East** corner (positivity)
- Main Entrance → **North or East** (wealth & health)
- Garage → **North-West** or South-East (movement energy)

9.3 ACCURACY

The performance of the Smart Real Estate Price Prediction system was evaluated using machine learning evaluation metrics after training the model on the dataset. The dataset was first preprocessed by converting categorical variables such as city, property type, furnishing status, and facing direction into numerical values using label encoding. After preprocessing, the dataset was divided into training and testing datasets using the train-test split technique. Around 80% of the data was used for training the model and 20% was used for testing the model performance.

A regression-based machine learning algorithm was used to predict the property price based on various housing features such as property size, number of bedrooms, number of bathrooms, floor details, amenities score, and distances to important infrastructure facilities. After training the model, predictions were generated on the test dataset and compared with the actual property prices to evaluate the model's accuracy.

The evaluation results show that the model performs effectively in predicting housing prices based on the given features. The model achieved an R^2 score of approximately 0.90, which indicates that around 90% of the variation in property prices can be explained by the features used in the model. This high R^2 value demonstrates that the selected variables have a strong relationship with the target variable and that the model is able to capture the patterns present in the dataset.

In addition to the R^2 score, the Mean Absolute Error (MAE) was also calculated to measure the average difference between predicted prices and actual prices. The MAE value indicates that the prediction error is relatively small compared to the overall price range of the dataset. This suggests that the system is capable of producing reliable price estimates for residential properties.

Overall, the accuracy results indicate that the Smart Real Estate Price Prediction system performs well in analyzing property features and generating price predictions. The combination of location factors, property characteristics, and infrastructure information helps the model learn meaningful patterns from the data. As a result, the system can assist buyers, sellers, and real estate analysts in estimating property values more effectively.

Accuracy Results

- Total dataset size: 2844 records
- Number of features used: 21 features
- Training data: 80%
- Testing data: 20%
- Machine Learning Algorithm: XGBoost Regressor
- Evaluation Metrics: R^2 Score and Mean Absolute Error
- Model R^2 Score: 0.92 (92% prediction accuracy)
- Mean Absolute Error: Low deviation between predicted and actual prices

CONCLUSION

10.CONCLUSION

This project successfully developed a machine learning-based system for predicting residential property prices across 12 major cities in Tamil Nadu, India. The system was built using a dataset of 2,844 property records with 22 features covering physical characteristics, geographic location, infrastructure proximity and qualitative attributes such as furnished status and amenities score.

The project followed a complete data science pipeline beginning with data collection and proceeding through data cleaning, exploratory data analysis, feature engineering, model training and evaluation. During the data cleaning phase, duplicate records were removed, missing values were imputed, inconsistent categorical formats were standardized and outliers were treated using the Interquartile Range method. Six descriptive analysis modules were implemented revealing key insights about the Tamil Nadu property market, including the dominant role of property area, the strong influence of distance to city center, the lifestyle premium associated with high amenities scores and significant city-level price variations.

Four machine learning regression algorithms were trained and rigorously compared using 5-fold cross-validation and a strictly held-out test set. Linear Regression achieved an R^2 of 0.68, Decision Tree achieved 0.74, Gradient Boosting achieved 0.87 and Random Forest achieved the best performance with an R^2 of 0.89 and a Mean Absolute Error of Rs. 3,20,000. The Mean Absolute Percentage Error of 8.5 percent indicates that the Random Forest model predicts property prices within approximately 8.5 percent of actual market value on average.

Feature importance analysis revealed that property area was the single most important predictor accounting for 31 percent of model predictive power, followed by distance to city center at 18 percent, amenities score at 14 percent, number of bedrooms at 11 percent and age of property at 9 percent. These findings confirm that both the physical characteristics of a property and its location and accessibility features are essential dimensions of residential property value in Tamil Nadu.

The progressive improvement from Linear Regression to ensemble methods demonstrated that property pricing involves complex non-linear relationships and feature interactions that simple models cannot capture. The Random Forest model's ability to learn these interactions automatically such as the city-conditional amenities premium and the property-type-dependent location premium was a key reason for its superior accuracy.

In conclusion, this project demonstrates that machine learning, when applied with rigorous data preparation, thoughtful feature engineering and systematic model evaluation, can deliver highly accurate and interpretable residential property price predictions that have genuine real-world value for all participants in the Tamil Nadu housing market.

FUTURE ENHANCEMENT

11. FUTURE ENHANCEMENT

The Smart Real Estate Price Prediction system provides an effective solution for estimating property prices based on various housing features and location factors. However, the system can be further improved in the future by integrating additional technologies and expanding its capabilities. Future enhancements can make the system more accurate, user-friendly, and useful for real estate buyers, sellers, and investors.

One possible improvement is the integration of real-time real estate data from online property platforms. Currently, the system works with a static dataset, but connecting the system with live property listing websites can help update the data regularly and provide more accurate and up-to-date price predictions. This will allow users to analyze current market trends and make better investment decisions.

Another enhancement can be the inclusion of more location-based factors such as crime rate, traffic conditions, nearby shopping centers, and environmental factors. These elements can significantly influence property prices and buyer preferences. By incorporating additional features, the prediction model can better understand the real estate market and generate more precise results.

The system can also be improved by implementing advanced machine learning and deep learning algorithms. Techniques such as Random Forest, Gradient Boosting, and Neural Networks can be used to compare performance and improve prediction accuracy. Using multiple models and selecting the best-performing model can further strengthen the system.

Another important enhancement is the development of a fully interactive web application where users can easily enter property details and receive price predictions instantly. Integrating the system with modern web frameworks and deploying it on cloud platforms can make the system accessible to a wider audience.

In the future, the system can be enhanced by including rural and semi-urban areas in the dataset. Currently, the model mainly focuses on urban property data, but by adding rural property information such as land prices, village housing data, and developing infrastructure details, the system can provide price predictions for a wider range of locations and support people who are planning to buy or invest in rural real estate.

Overall, these future enhancements will help transform the Smart Real Estate Price Prediction system into a more powerful and intelligent real estate analytics platform that supports better decision-making in the property market.

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